the magazine for
the contemporary teacher

TEACHER PLUS

March 2025
₹ 80

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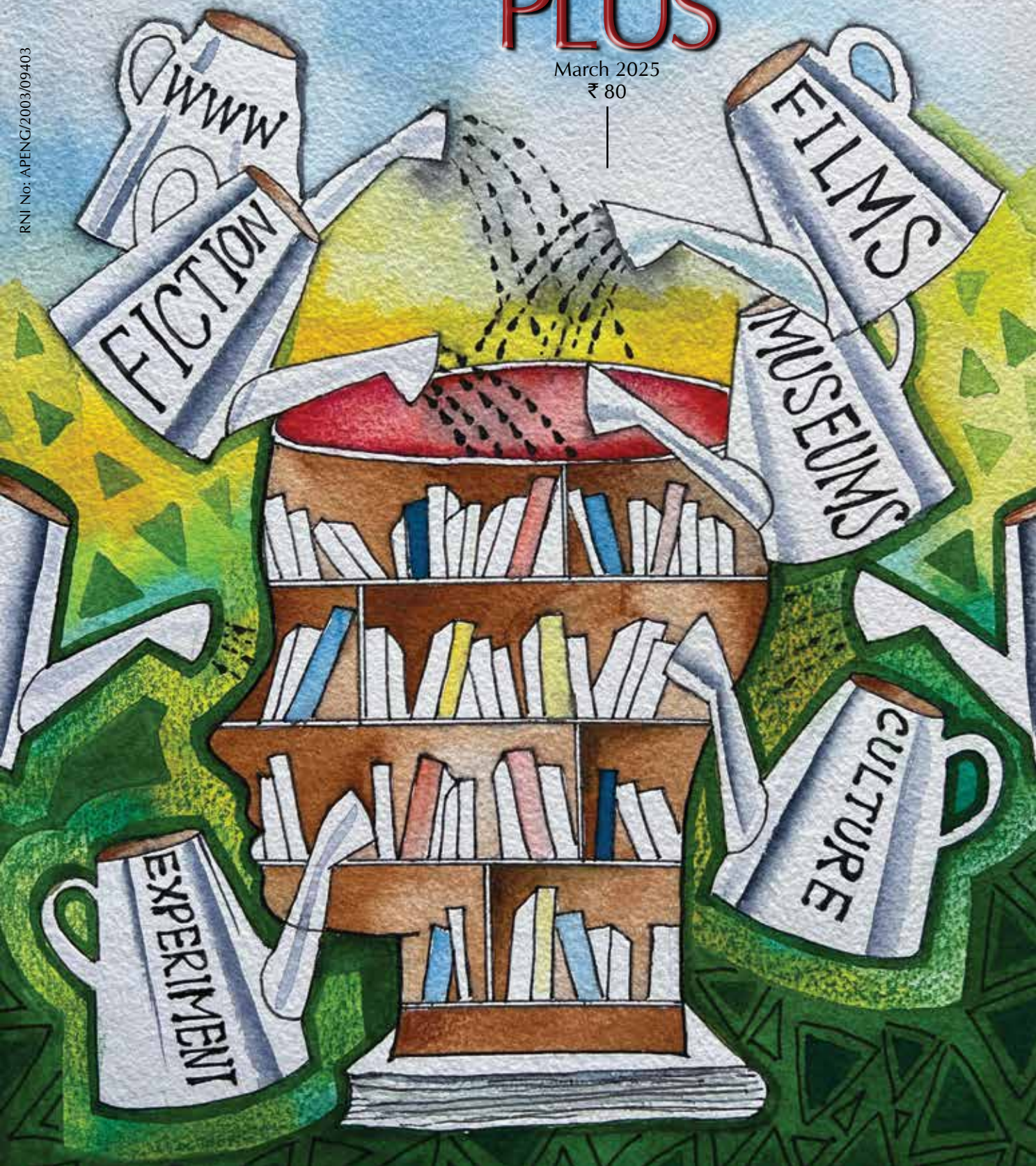
RNI No: APENG/2003/09403



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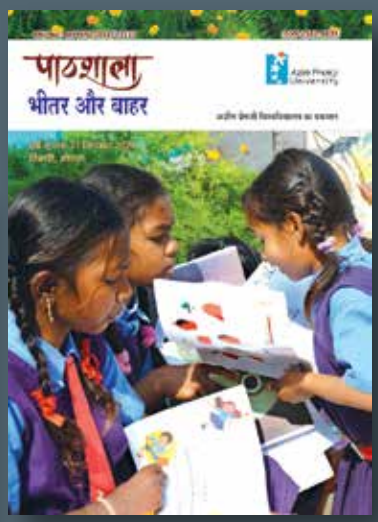

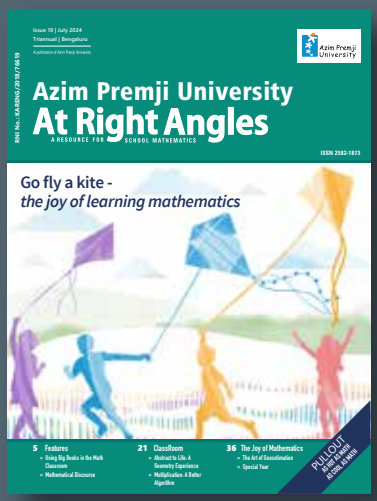

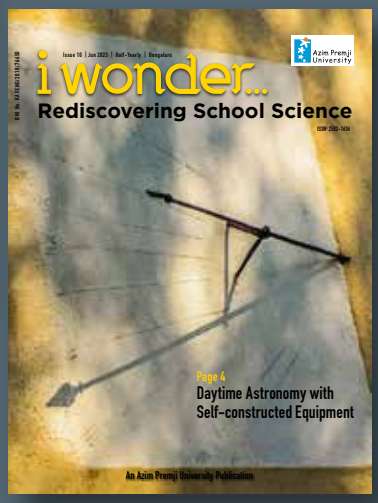

contents

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Editorial 5

A Step Ahead
Stopping the [doom]scroll 6
Neerja Singh

Cover Story
We often think of the entire curriculum being delivered on the shoulders of a set of textbooks in each subject, but in reality, teachers draw on a range of other resources to help children make sense of concepts and themes. These resources allow us to develop a more contextualised understanding of the content, and in higher classes, also be introduced to the idea that knowledge develops within a social, political and cultural structure and can change over time. Textbooks are designed to convey the minimum essential information on a topic, often with prompts that can take the learner to other resources to expand and fill out the larger picture. These resources can be fiction, additional non-fiction readings, films, museums and the whole range of materials available on the internet. In this issue, we tease out the idea of context-providing materials of various kinds, which can encourage learners to explore topics beyond the limited – and sometimes skewed – range of information that is available in the prescribed textbook. 9 - 17

Teaching with impact: the power of resources
Nidhi Solanki

These books are made for learning
Subha Das Mollick

Ecowatch
Living with microbes 18
Nandini Dholepat

First Steps
Exploring the world through children's eyes
Chintha Keerthana 20
Children's art isn't just shapes and colours; it's a reflection of their world. Chintha Keerthana writes that we ought to listen to their interpretations to understand the many ways to be creative.

Reflective Practice
Seeing 'Missed Takes' in mistakes 22
Neeraja Raghavan

Notes from a Teacher's Diary
From resentment to empathy
Khrawkupar Kharshiing
24

Project
Mountains of lessons
Mala Kumar
What if every glass of water you drank had a story that began in the mountains? Mala Kumar's interdisciplinary classroom project will help children understand why mountains matter.
26

The 7Cs
Creating responsible citizens
Aruna Sankaranarayanan
31

The Other Side
Nurturing talent beyond academics
Anuradha C
What if your child's true talent lies outside textbooks? Anuradha C writes about how parents and schools can nurture children's skills in sports, music, and arts without compromising essential learning.
34

Comment
Implementing global citizenship education in India: challenges and considerations
Dr. Gh Hassan Wani
37

Math Speak
Getting ready to take on the world
Monica Kochar
Climate change, global health, and math: Monica Kochar writes about real-world applications that make math learning both engaging and meaningful.
40

The Hope Series
Cultivating hope in the classroom
Jane Sahi
42

Things to Think About
Dear teacher, I have so much to say
Bhuvaneswari B and Sapna Bhat
What if a struggling student could tell you exactly what they need? Bhuvaneswari B and Sapna Bhat pen a letter addressed to teachers from a student with ASD and ADHD, sharing his hopes, challenges, and dreams.
46

Focus
From the 'world' to the 'word': a Freirian look at reading
Amruta Veer and Swati Jain
49

Cogitations
The Constitution and General Will of people
Prakash Iyer
How did 300 people decide India's democratic fate? As the conversation between Ira and Anubhuti continues, Prakash Iyer shares how the Constitution shaped our society.
52

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Teacher Plus is supported by Azim Premji University

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Registered with the Registrar of Newspapers of India under RNI No: APENG/2003/09403

Unsolicited submissions are welcome. Please address all correspondence to
Teacher Plus
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ISSN No 0973-778

Vol. 23, No. 3; Pages 56

Cover: Re'sources' of knowledge

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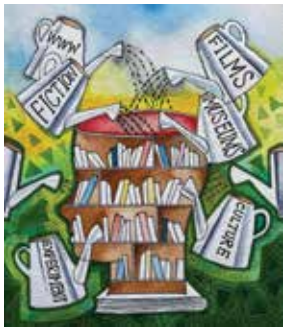
Text in context



For a few years now, our columnist Neerja Singh has been writing about how adults can stay "a step ahead" of the children we interact with, particularly in relation to digital tools and their use. Her column acknowledges the challenges of keeping up with accelerating changes in our environment, and the confusions that can arise when we see children occupying a world that is so different, in so many ways. While there's no doubt that it is hard to find the right balance in parenting in these times, the teacher's task poses particular difficulties. The teacher must not only keep up with modifications to curricula and shifts in pedagogic practice, but also be attuned to the way culture morphs from one generation to the next. And as one grows older, there is the feeling that you're never quite going to be able to stay abreast, let alone get ahead of the changes. Yet, one must find a way to build bridges with the young, to find channels of communication that allow us to exchange our ideas, trading learning gathered by the adults from experience over years for that learning gathered by the young in the moment.

I recently asked my students to build a lexicon of their times, to list and explain new words that described their interactions with each other and the world, as well as actions they performed with their digital devices. In the space of half an hour, I acquired a vocabulary of several new words, from a new way of understanding "aura" to learning that one can "yank" text from one source and place it somewhere else. And of course that word that made it to this year's Oxford English Dictionary Word of the Year – brain rot. This exercise reminded me of a piece of advice I had come across early in my teaching career, that one must always begin in the world of the child. What does the child know and how does the child interact with her immediate environment? This advice forces us also to acknowledge the child's context of learning, and living, and attempt to make connections between the text (curricular material) and the context (the child's life and surroundings). Starting my lesson by asking the students to draw on their own insights, developed through their own observations, led not only to engagement, but also to a sharper attention to their immediate environment – and to learn from this reflection.

Our cover theme for this issue pushes this idea further, taking it in two different directions: Nidhi Solanki talks about how we can bring the world into our classrooms, at different levels. And Subha Das Mollick offers a potted history on a very special set of readers that attempt to open up a child's mind. There are many ways to help contextualize learning for the child, and we'd love to hear how our readers might approach this question, But whatever the approach, for the teacher, it can help not only go *a step ahead*, but more importantly, to stay *in step with* the child and her learning journey.

Usha Raman

Stopping the [doom]scroll

Neerja Singh

Illustrations: Shilpy Lather

In the ever-evolving world of technology, one of marketing's latest mantras is "stop the scroll." This credo highlights the reality of how deeply scrolling is ingrained in our daily lives. Whether it's social media, short-form videos, or news feeds, the act of scrolling has become a reflex, a habit tied to our thumbs, and increasingly, to our minds. While advertisers see this as an opportunity to grab attention, educators must grapple with the phenomenon of "doomscrolling" – a behaviour that has individuals compulsively scrolling through negative or distressing content online. This issue is especially concerning for schoolchildren, who, instead of engaging in healthy learning or social activities, are consumed by endless scrolling. The implications for mental health, focus, and overall wellbeing are profound.

To understand the impact of scrolling on children, it's essential to unpack the changes it brings to the human mind. Research on attention spans, although contested, consistently suggests a decline. Some studies claim the average attention span has reduced from 12 seconds to as low as eight seconds in the social media age. Regardless of the exact number, the conclusion is clear: the sheer volume of content and its algorithmic

delivery have reshaped how young minds process information. Social media platforms are designed to provide endless personalized feeds, which while engaging, can also be overwhelming. For children, this poses unique challenges. Their brains are still developing, making them particularly susceptible to the dopamine hits that come with every scroll, like, or share. This constant stimulation can make it difficult for them to focus on tasks requiring sustained attention, such as reading, problem-solving, or participating in class discussions. Moreover, the personalized nature of feeds means they're often exposed to echo chambers or distressing content tailored to their fears or insecurities, feeding into the doomscrolling cycle.

Doomscrolling refers to the compulsive consumption of negative news or content, often leaving individuals feeling anxious or helpless. For children, this behaviour is particularly concerning. A newsfeed filled with climate change disasters, political unrest, or personal insecurities amplified by social media comparisons can contribute to feelings of anxiety, depression, and eco-anxiety. These emotions, when unchecked, seep into classrooms, affecting academic performance and peer relationships. There is a growing trend of students discussing catastrophic events they read about online, from wildfires to global pandemics. These conversations may not be rooted in classroom lessons but may stem from the headlines they encountered during late-night scrolling. There has been an uptick in students expressing fears about the future after spending hours consuming content about economic instability and climate crises.

Scrolling is not inherently bad. It offers access to a wealth of information, sparks creativity, and provides new ways to connect. However, the medium often dictates the message. Short-form content, designed for quick consumption, prioritizes entertainment over depth. For students, this can lead to surface-level engagement rather than deep learning. Moreover, the fragmented nature of scrolling affects how children process information. They're trained to skim rather than analyze, making it harder to develop critical thinking skills. The constant influx of new content leaves little room for reflection or synthesis, key components of meaningful education. Teachers often find themselves competing for attention in a world where every piece of information is designed to "stop the scroll." There is a growing struggle to focus on long-form reading assignments in class. Many admit they are accustomed to skimming social media posts and find it difficult to engage with dense texts.

While schools cannot shield students entirely from the digital world, they can equip them with tools to navigate it in a healthy way. Teaching students to critically evaluate the content they encounter online is a vital step. Discussing how algorithms work and why certain content appears in their feeds can help them distinguish between credible sources and sensationalized material. By understanding the mechanics behind social media, students can become more discerning consumers. Introducing a digital literacy module in school where students analyze their social media feeds, identifying clickbait headlines, and exploring how algorithms tailor content to individual users, would be a first step. This exercise would not only educate them but also spark conversations about responsible consumption.

Introducing students to the concept of mindfulness in their digital habits is another crucial strategy. Encouraging them to set boundaries, such as designated screen-free times or breaks from social media, can make a significant difference. Tools like screen time trackers or apps that limit usage can be helpful. Mindfulness practices like meditation or journalling can also counterbalance the constant stimulation of digital devices. Keeping a "scrolling diary" for a week, documenting how much time they spent online and how it made them feel would likely be an eye-opening exercise for both students and teachers.

Providing a platform for students to share their thoughts and feelings about the content they encounter online can also help. Classroom discussions or counselling sessions allow them to process their emotions and understand that they're not alone in feeling overwhelmed. Weekly "Media Monday" sessions, where students could discuss news stories or trends they had seen online would not only build critical thinking skills but also help students articulate their concerns and support one another.

Counteracting the effects of short-form content by incorporating activities that require sustained focus is essential. Encouraging students to read books, engage in debates, or work on projects that demand critical thinking and creativity highlights the value of depth over breadth in learning. Replacing traditional tests with project-based assignments, asking students to research historical events and present their findings is sure to lead them to deeper engagement and collaborative learning. Teachers can also lead by example, demonstrating balanced digital habits and sharing their own strategies for managing screen time

or finding uplifting content. When students see adults navigating the digital world responsibly, they're more likely to follow suit.

Rather than banning devices, integrating them into the learning process in constructive ways can show students that technology can be a force for good. Educational apps, collaborative platforms, or digital storytelling tools can highlight stories of innovation and resilience to counter the negativity in their feeds.

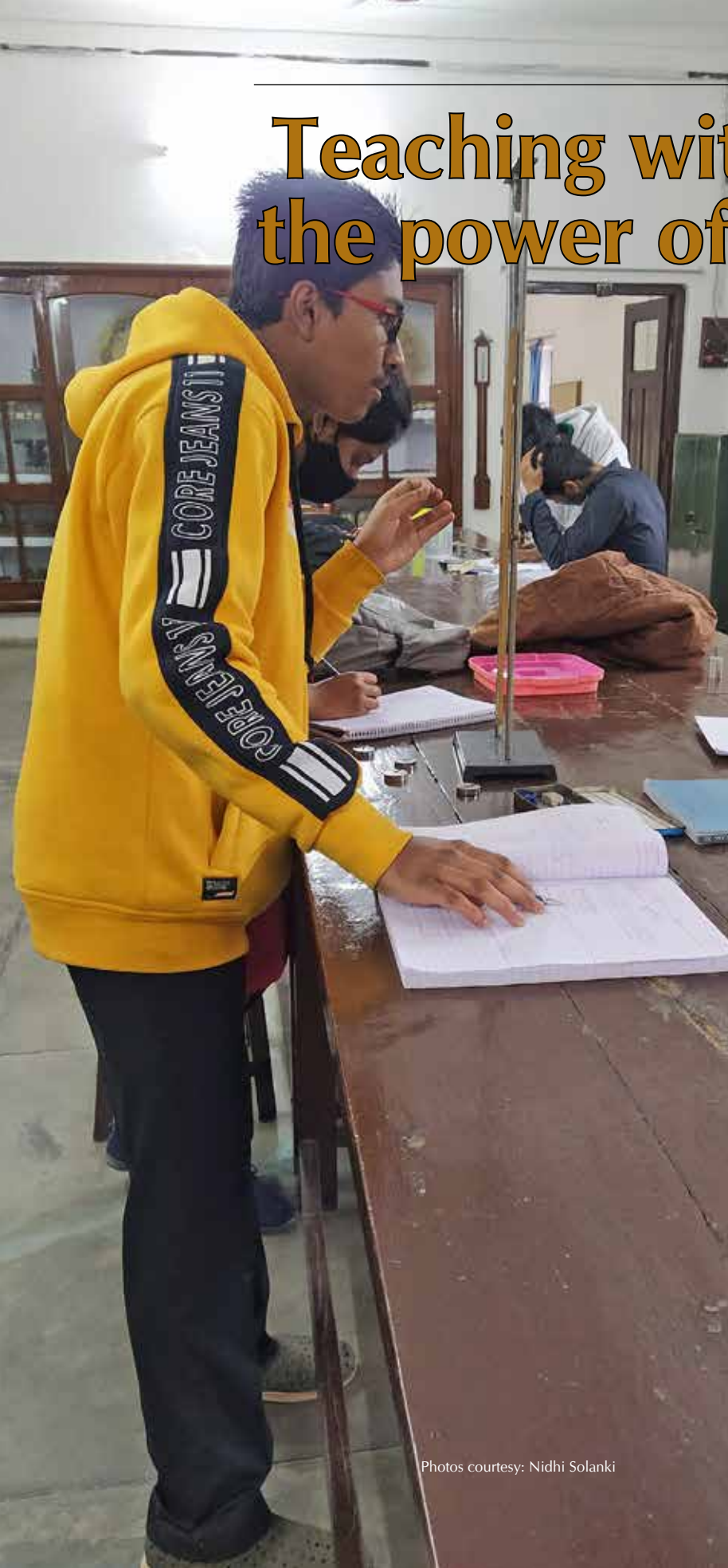
Teachers cannot address doomscrolling by themselves. Collaboration with parents and the broader community is essential. Schools can organize workshops for parents to recognize and address doomscrolling behaviours. Community programs can provide resources and support for children struggling with digital addiction or anxiety. Parents, in particular, play a crucial role. They can monitor their children's online activity, set household rules for screen time, and engage in conversations about their digital experiences. Encouraging offline activities, such as sports, arts, or volunteering, can provide children with alternative sources of fulfilment and joy.

In a world designed to keep us scrolling, teaching children to "stop the scroll" is an act of empowerment. By addressing the phenomenon of doomscrolling and promoting healthier digital habits, teachers can help students reclaim their focus, foster critical thinking, and build emotional resilience. While the digital age brings challenges, it also offers opportunities to rethink how we engage with information and each other. As educators, the goal is not to resist technology but to guide students in using it wisely, ensuring that scrolling becomes a tool for growth rather than a trap of negativity.

Neerja Singh is a generational diversity speaker, author of six generational books, and a consultant on working with GenZ; Leading with Social Impact, and Suicide Prevention and Destigmatisation. A millennial in spirit, with the benefit of hindsight, she uses her 37 years in media and education to help leverage generational diversity at work and at home. She can be reached at https://www.linkedin.com/in/neerja-singh/.

Teaching with impact: the power of resources

Nidhi Solanki



Photos courtesy: Nidhi Solanki

It was a bright, crisp morning at 10 am when I pedaled my bicycle toward the senior building of the school. A routine day turned into something extraordinary as I noticed my fellow teachers congregating in the garden – the designated assembly point for that day. When I reached the classroom, I asked my 11th grade students a question: "How does a bicycle function?" This simple inquiry opened the floodgates of curiosity. Over the following weeks, our classroom transformed into a hub of discussions about motion in physics. Students dissected concepts like rolling and sliding friction and explored the forces that set a bicycle in motion. This single question set the stage for an immersive learning experience, where an entire topic unravelled with nothing more than a bicycle and a group of eager learners.

At the heart of teaching lies a fundamental question: what role does a teacher play in learning? Learning, after all, is an innate process that occurs naturally, with or without a teacher's presence. The teacher's role, then, is to nurture this process, cultivating an environment where learning can flourish. Within this framework, two schools of thought emerge. One advocates for children choosing their learning topics, thriving in spaces like Summerhill, Aarohi Learning Centre, and BeMe in Bengaluru. The other follows a structured curriculum crafted after

extensive research into a child's developmental needs. Both approaches have their merits, but this article does not aim to debate their efficacy. Instead, it focuses on the teacher's role in orchestrating meaningful learning experiences. Whether students present topics to their teacher-facilitator or educators independently select subjects, success hinges on the teacher's preparation and execution.

Shifting paradigms in teaching

In olden times, teaching was predominantly lecture-based, often characterized by a single teacher and a textbook-driven approach. While this method has its merits, contemporary education demands more dynamic, engaging strategies. Relying solely on lectures is no longer sufficient to meet the diverse needs of today's learners. Across my decade-long teaching journey, I have experimented extensively to discover what resonates with young minds and what doesn't. Contrary to the notion that educators stick to one teaching method in their entire career, many of us invest countless hours, meticulously preparing for each chapter. This preparation involves choosing the right resources to facilitate learning – a process that is both critical and creative.

Why diverse resources matter

The need for diverse resources stems from more than just the desire to make learning interesting. Different resources cater to varied learning styles, allowing each student to engage with the material in a way that suits them best. Additionally, diverse resources enable deeper exploration of topics, encouraging students to think critically and holistically. The nature of the subject often dictates the type of resources required. For example, physics encompasses both abstract concepts and practical applications, necessitating a blend of empirical methods, hands-on activities, and reflective writing.

Using diverse resources also brings a sense of newness and excitement for both students and teachers. Currently, I am on a break, but when I was teaching, I found the process of preparing and incorporating various resources invigorating. Some chapters allowed me more preparation time than others, and I relished the opportunity to explore websites, books, and subject specialists to design engaging lessons. This preparation often left multiple paths open, allowing learning to take its own organic shape.

Collaboration and reflection in teaching

Preparation, however, is not without its challenges. At times, I encountered difficulties during the planning or execution of lessons. In such moments, I sought help from colleagues, subject experts, and even friends outside my field. Their feedback often provided fresh perspectives, helping me simplify complex concepts and make them relatable.

Keeping explanations simple and jargon-free became a cornerstone of my teaching philosophy. Many students entered my classroom with preconceived notions about physics being a difficult subject. By connecting concepts to real-life and making lessons approachable, I aimed to change these perceptions.

Inside the classroom: electricity as a case study

A topic I taught across multiple grades was electricity. My approach varied depending on the grade level. For grade 8, the focus was on hands-on activities, with nearly 20 classes dedicated to experiments using simple materials like bulbs, wires, and batteries. In grade 10, the sessions used to strike a balance between hands-on activities and conceptual discussions, while grade 12 emphasized problem-solving and advanced equipment like capacitors and LEDs.

Before each session, I ensured the availability of materials needed for the experiments. Allowing students to perform the activities themselves was crucial, as it encouraged discovery and collaborative learning. For instance, during a lesson on circuits,

students built their own circuits, asked questions, and engaged in discussions. These discussions often led to deeper understanding, as students challenged each other's ideas and clarified concepts collaboratively.

Simulations also played a significant role in teaching electricity. Platforms like PhET allowed students to explore circuits and observe results instantly, complementing the hands-on experiments. The physics lab became an extension of the classroom, offering access to equipment for demonstrations and experiments like Ohm's Law, PN junction diodes, and optics setups.

Resourcefulness in government schools

I have worked in government schools as well, and this often required creative solutions due to the limited availability of resources. For example, when a bob was unavailable for an experiment, I used round stones as substitutes. While this led to slight deviations in values, it did not hinder the understanding of core concepts. Similarly, while teaching density to grade 8, I have used kitchen items like vegetables and fruits to explore buoyancy. Students brought various objects, tested them, and formed hypotheses based on their observations. This openness to exploration fostered a sense of curiosity and creativity.

Learning beyond the classroom

Some of my most impactful lessons took place outside the classroom. For example, while teaching the laws of motion to grade 9, we calculated average speed using a marked ground and a timer. This hands-on activity ensured that students would never forget the concept of average speed. Similarly, while teaching waves, we visited a swimming pool and a nearby pond to observe wave production formed by throwing a stone and the interaction of the disturbance created by two stones. The visual impact of seeing waves meet and interact left an indelible impression on the students.

Projectile motion lessons also incorporated real-life data with students comparing theoretical and actual ranges and angles. Working in small groups, they discussed how factors like air resistance and technique influenced the results. These outdoor

activities not only made learning memorable but also demonstrated the relevance of physics in everyday life.

Leveraging external resources

In addition to hands-on activities, I utilized videos of experiments and lectures by renowned educators like Walter Lewin. Inviting subject experts to the classroom added another layer of depth to the learning experience. For instance, I had a biology teacher explain a concept (cell structure and function) to grade 8 students, while an engineering department head discussed material properties with senior students. A physics researcher's guest lecture further enriched the learning process.

Peer teaching also became a valuable tool in my classes. Students often explained topics to one another, using creative methods that fostered confidence and collaboration. With younger students (ages 5-11), we built simple models like balloon rockets and parachutes, using resources from the Arvind Gupta Toys website. These activities combined play and learning, making complex concepts accessible to young minds.

Writing and reflection

Encouraging students to write and reflect on their learning used to be another critical aspect of my teaching. Instead of copying notes from the board, students were encouraged to process their understanding and express it in their own words. This approach often involved creativity, with students incorporating diagrams, sticky notes, and varied formats into their work.

The role of technology

Online resources like simulations, video lectures, and educational websites became integral to my teaching. Depending on the group size, I used projectors or laptops to show videos, articles, or presentations. Rarely relying on traditional lectures, I embraced multimedia tools to create engaging and interactive sessions.

Beyond physics: debates and games

Activities like debates and games also found a place in my classes. For instance, we debated topics like nuclear energy, not to reach a single solution but to explore diverse perspectives and challenges. These activities encouraged critical thinking and collaborative problem-solving. Treasure hunt was another creative way to include these skills. It's a game where you move to the next level if you unravel the question or hint shared in the first level, as you look for a certain 'treasure'. The learners themselves organized the game, where a couple of them prepared the questions and hints. It was so much fun. After these games, we would discuss the experience and learning.

Books and assignments

Books are like companions who help and guide you. I referred to globally recognized books like Resnick Halliday, Paul G. Hewitt, and Feynman Lectures to enrich my lessons. Assignments were research-based, often requiring students to explore both fiction and non-fiction resources. Whether working individually or in groups, students delved into topics using diverse resources, fostering a culture of inquiry and exploration.

Addressing diverse learning needs

Some of my classes had students with diverse learning needs. While I addressed some of these needs myself, I collaborated with special educators and psychologists to design customized plans and activities for others. This collaboration was vital in ensuring inclusive learning, creating a classroom environment where every student could thrive. Together, we explored various strategies to cater to these needs, such as providing tailored worksheets or organizing separate classes when necessary.

Final reflections

Teaching is a dynamic, ever-evolving journey. It is not just about imparting knowledge, but about inspiring curiosity, fostering creativity, and building connections. By using diverse resources and approaches, I aimed to make physics not just a subject but a lens through which students could view and understand the world. As educators, our ultimate goal is to empower students to become lifelong learners, equipped with the tools to explore and question the world around them.

However, the most important aspect at the heart of all this is the *relationship* with learners – understanding their needs and getting to know them as individuals. Often, teachers give their best, yet they may see little to no learning happening. This could be because the learner is facing difficulties at home, with friends, or with the subject itself, or perhaps they are dealing with personal challenges. The foundation of effective learning lies in building a strong, trusting relationship. Without it no meaningful learning can take place.

My own trajectory in school changed because of a teacher. I used to be terrified of math, but this teacher made me feel so comfortable that the subject began to make sense to me. It was his motivation, support, and unwavering belief in me – never putting me down, instead seeing me as capable and important – that changed everything. That's the power of a teacher, and it shows how crucial this role can be.

Years later, you come to realize how profoundly you've impacted someone's life for the better. Without a strong teacher-student relationship, no amount of resources or infrastructure can truly make learning happen.

Nidhi Solanki has taught physics at alternative schools like Anand Niketan Democratic School, Rajghat Besant School KFI in Benaras, and Prakriti School in Noida, using creative inquiry-based learning methods. She has also collaborated with NGOs such as Ekalvya and the Azim Premji Foundation. For the past year, she has been delving into various realms of education, examining what truly shapes a learner's experience. She can be reached at <solankinidhi34@gmail.com>.

Form IV

Statement about ownership and other particulars about newspaper (Teacher Plus) to be published in the first issue every year after the last day of February.

1. Place of publication : A 15, Vikrampuri Secunderabad 500 009
2. Periodicity of its publication : Monthly
3. Printer's Name : Usha Raman
   Nationality : Indian
   Address : A 15, Vikrampuri Secunderabad 500 009
4. Publisher's Name : Usha Raman
   Nationality : Indian
   Address : A 15, Vikrampuri Secunderabad 500 009
5. Editor's Name : Usha Raman
   Nationality : Indian
   Address : A 15, Vikrampuri Secunderabad 500 009

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Usha Raman

01-03-2025 Signature of Publisher

These books are made for learning

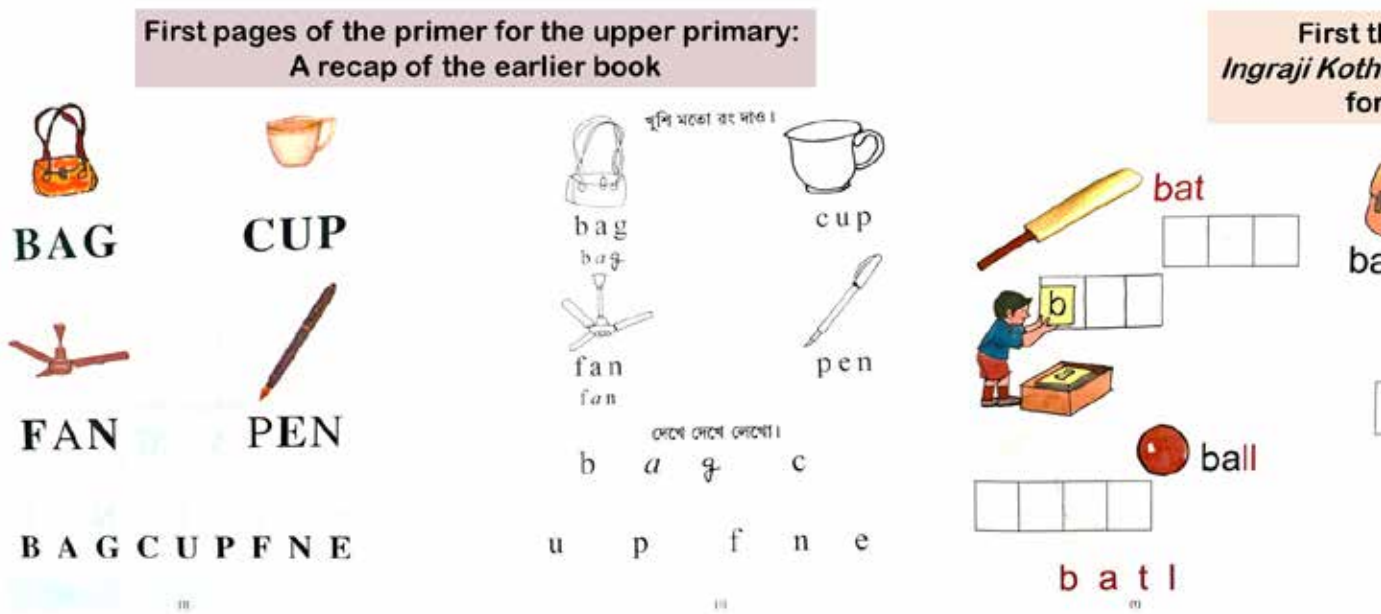
Subha Das Mollick

One of the happiest days in every child's life is the day she gets new textbooks in a new class. She smells the shining new books, lovingly covers them and proudly writes her name. Thus begins her yearlong engagement with these resources of knowledge. By the end of the year, the textbook is dog eared, the margins get filled with notes, and passages get underlined with 'Impt' written beside them. The more studious the child, the more are the marks of engagement with the book.

Educationists and producers of textbooks are aware of the importance of textbooks in a child's life. They try to contextualize the textbook to the child's lived experience – and this is a big challenge for a country like India. NCERT brings out textbooks aimed at the entire nation and they face the challenge of making the texts relatable to children growing up in a diverse cultural milieu. Months of brainstorming sessions precede the writing of every NCERT textbook.

Of all textbooks, primers hold a special position because they are a gateway to the world of letters for children. Doyens of Bengal Renaissance like Ishwar Chandra Vidyasagar and Rabindranath Tagore have channelized their creative energy into writing primers like *Barna Parichay* and *Sahaj Path*, which *has beautiful linocut graphics illustrated by the highly acclaimed* artist Nandalal Bose. Both *Barna Parichay* and *Sahaj Path* are primers for Bengali. What about English primers?

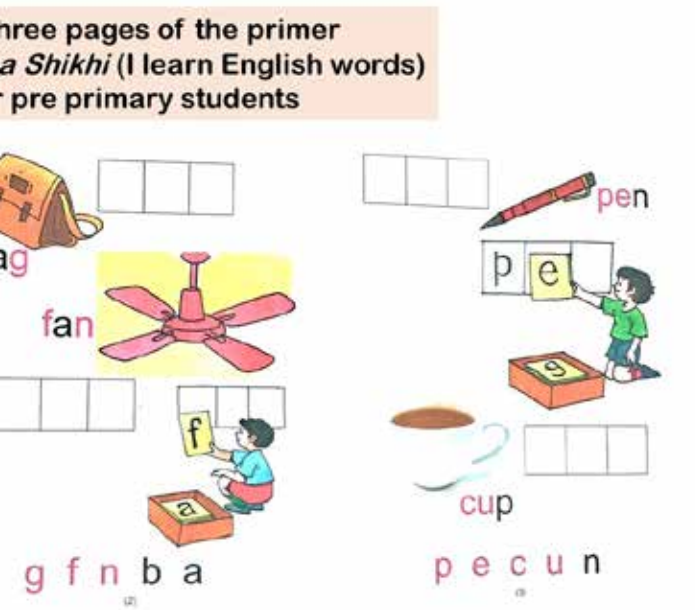
Debabrata Majumdar, a professor of physics at a college in West Bengal, developed a keen interest in primary education at the turn of the new millennium and started an organization called "Shekhar Sathi", meaning "Companion in learning". In 2003 he published a Bengali primer called "Amar Boi" (My book).Close at heels followed the first English primer. These primers gained significance after the publication of the first ASER report in 2005, which revealed the abysmal state of basic literacy and numeracy in India.



Images courtesy: Subha Das Mollick

Faced with the challenge of contextualizing English learning taking into accounts the lived experience of a child growing up in a village and not exposed to English, Majumdar decided to take a novel path. He began by making a list of English words in the Bengali vocabulary. Words like bat, ball, bag, van, taxi, train, etc., figured in this list. Using these words, he developed his first primer meant for children aged five. The first two pages of this primer are illustrative of his method. It is clear that he has deviated from the convention of 'A for Apple, B for Ball'. Instead, he has matched familiar objects with their English names. The child sees the word 'bat' and, in his mind, associates the word with the object. But the word consists of letters. How does one make the child aware of these letters as independent entities? In the next word 'ball', the child encounters two common letters and two new letters. The new letters are highlighted in red. Thus, on page 1, through pictures of familiar objects, the child starts to recognize four letters of the English alphabet. These letters are written at the bottom of the page. Again, on page 2, there are two familiar objects and two words associated with these objects. These objects have some familiar letters and some new letters. All the letters are listed at the bottom of the page – new letters in red and familiar letters in black.

By page 13, the child gets acquainted with all the 26 letters of the English alphabet. On page 14 he writes these letters inside assigned boxes – all in lower case. On pages 15 to 19, the child sees more pictures with their associated words. Some of these words



are part of the Bengali vocabulary, some are not. For example, on page 15, table and line are familiar words, but book and boy are not. So, the child learns new words.

From page 20 onwards, numbers are introduced – numbers spelled out as well as numbers as numerals. By page 24, numbers 1 to 10 are introduced. Pages 25 to 28 are drawing exercises. The child has to read the word and draw its picture in the box above the word.

In the note to teachers at the beginning of the book, there are clear-cut instructions on how to get the best out of this book. Two points are noteworthy in this set of instructions. First, the teacher should sit with small groups of children. Group learning is always more enjoyable and faster. Secondly, the teacher should never try to underline any rule or logic behind the use of letters to form words, or the use of singular and plural in case of numerals.

Part II of the primer, meant for children aged six, begins with a recap of the words they already learnt the previous year – but this time, the words are in upper case letters. So the child recognizes the object from its picture and gets introduced to upper case letters – a continuity from the known to the unknown. On pages 2 and 3, there are outlines of the objects for the child to fill with colour, along with the words written in upper case and lower case. At the bottom of the page, all the letters used in the words are listed and the child is asked to copy the letters. The point to be noted is that the instructions are all written in Bengali. This pattern continues in the next two pages, with some new objects thrown in. The new letters in the words are highlighted in bold and all the new letters are listed at the bottom of the page. Thus, the child is familiarized with all the 26 'capital' letters of the English alphabet. Something interesting happens on pages 8 and 9. Objects are presented in pairs – glass and dish, pant and belt, bat and ball, table and chair. Incidentally, all these words are commonly used in the Bengali vocabulary. So the words are not new to the child. Only, the English way of writing these words is new. In the next few pages, pictures of complete scenes are presented, in which the children can spot and label familiar objects.

Innovative exercises, in which the child has to complete the word by filling in the missing letter or writing the word that matches with the picture, will keep the child engaged and his leaning will go deep. The last page of the book gives a list of all the new

words learnt, arranged in alphabetical order. The words are also spelt out with Bengali letters, which will help the child with his English pronunciation. Some words are three letters long, while some are six to seven letters long. The logic behind the spellings of words like school, or chain, or cycle is not mentioned at this stage.

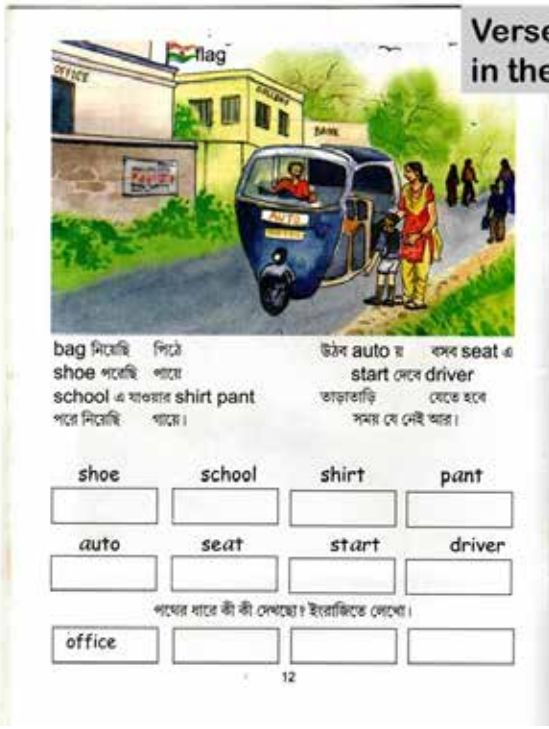
In the note to teachers, it is written that the child must feel confident that he can learn a foreign language. He will get this confidence by reading correctly and doing the exercises correctly.

How are Debabrata Majumdar's primers different from other primers? First, he does not go in the alphabetical order from A to Z, because the order is not really important for language learning. But in the two books, after the child gets acquainted with all the letters, he reads and writes all the letters in the English alphabet in order. So, the child knows the order of the 26 letters in the English alphabet. This would be useful to him later in life. Secondly, instead of choosing words that begin with particular letters, he has chosen words that the child uses in his everyday language. Most importantly, instead of going from alphabet to word, he has gone from word to alphabet. Also, he has not shied away from introducing big words at the pre-primary stage because the child is familiar with these words.

Sajal Jena, a primary school teacher at a village school in Hoogly district of West Bengal, shares his experience of using these primers. He says that he tries to bring the objects – ball, bat, bag, etc., to class and spends a lot of time with each object. The children, usually three year olds, play with the ball, examine its properties and talk about their own balls before the word 'ball' is introduced to them. The word is constructed using small pieces of cardboard bearing the letters. First the teacher constructs the words and then the children are asked to pick up the right cardboard pieces and construct the same word. This way, more than one day may be spent on one single page of the primer, resulting in unhurried, joyful, stress-free learning. Along with the child, the child's parents also pick up the English language. However, Sajal Jena laments that teachers prefer to stick to conventions and are too lazy to switch over to this new method.

Raju Baidya, a teacher in Bashirhat district of West Bengal, is another regular user of Debabrata Majumdar's primers. His daughter, who is now in Class III, has grown up on these primers. The best thing about these books is that new information is given in small doses. So the child is comfortable and gains confidence. The child can also use these primers without the guidance of a teacher. The instructions are simple and in Bengali.

Verses in mixed language in the primer for Class I



The same poem in English and Bengali in two different primers

Debabrata Majumdar has written six English primers – two for pre-primary classes and four for primary classes. The primer for class 1 has Bengali rhymes with English words – like the ones shown below. Then there are exercises in which the child has to write the names of four vehicles or four things to wear. These are advanced exercises in which he has to scan his memory and pick up the words that mean different vehicles from all the words he has learnt. Then there are pages dedicated to themes like school, hospital, or circus. Words related to this theme are presented in mixed language verses. This 40 page book can be comfortably completed in 120 days spread over an academic year.

The book meant for class IV children begins with a charming rhyme "Naughty Boy". Incidentally, the same rhyme is present in an earlier Bengali primer in Bengali. So the child feels happy and confident when he encounters a familiar picture. Debabrata Majumdar says, "All the texts of the class IV primer are contextualized texts of previous English/Bengali books. Identical illustration, somewhat similar texts, bring a smile in the child's face and a feeling 'I will read it easily'." This inter-textuality is interesting. It is a novel way of contextualizing a textbook.

In this book there are full passages in English that the child is expected to read and understand. The pictures and stories from the class 1 book are repeated here, but now the stories are completely in English, instead of a mixed language. The meanings of the new words are written in Bengali at the bottom of the page. So, with the help of his mother tongue – the language he speaks and hears at home – the child will earn proficiency in a foreign language.

Debabrata Majumdar's books are not meant for pan India use. These are English books written in Bengali. The names of the books are also in Bengali instead of English. The books are contextualized to the life of a child growing up in rural or suburban Bengal. However, similar books can be designed for children growing up in different villages spread across India. All Indian languages have English words in their lexicon and that is the starting point for developing these primers. Presently, Debabrata Majumdar is working on an English primer for Santhali children.

Note: *Dr. Debabrata Majumdar, author of the English primers, may be reached at* majumdar.debabrata@gmail.com.

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Living with microbes

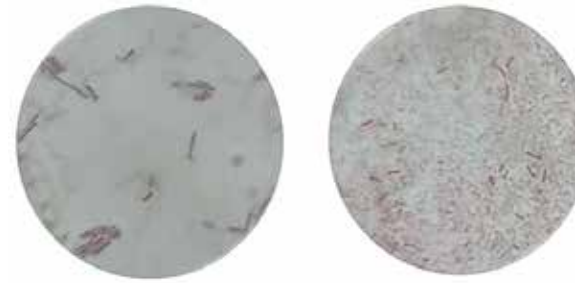
Nandini Dholepat

After the excitement of seeing fungi under the microscope, all of us were waiting to see bacteria. Jayanthi called us to a desk. "Come here and observe these slides," she said.
As we walked over to Jayanthi's desk, Sujatha noticed something and asked, "Jayanthi, why are you keeping this spoiled carrot here?"
"We use the spoiled carrot to prepare bacterial culture. Look through the microscope. Can you see the orange spots? These bacteria are long, rod-shaped and are called 'Erwinia'. They also attack other vegetables like potato, tomato, and cucumber," Jayanthi said.

Photos: Nandini Dholepat

"It is plant bacteria. What are the other kinds of bacteria we commonly find?" Rini asked.
"Lactobacillus is from milk. I diluted and strained some buttermilk to show it to you. Lactobacilli are among the most common probiotics found in foods such as curd. They help us maintain our health, and can treat diarrhea and skin disorders such as eczema."

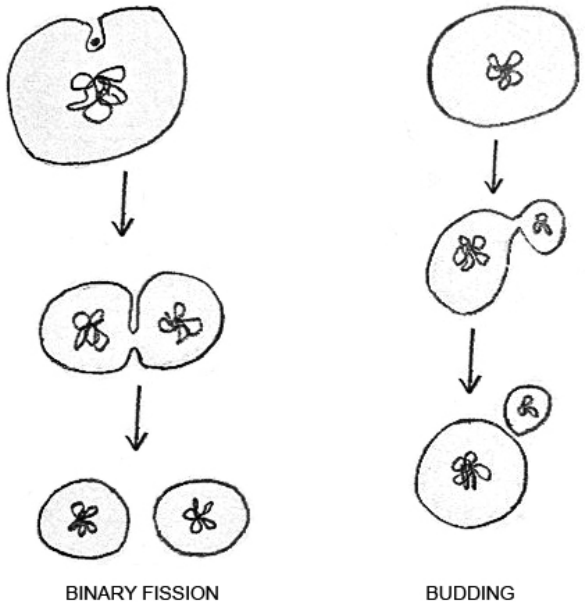
"Oh! I bought probiotic bottles from the market. Now I know why our elders ask us to consume buttermilk or curd regularly," Madhavi said.
"Have you heard of 'Escherichia coli'or 'E. coli'? It is part of the microbiota of our gut. It is normally found in the lower intestine of animals, including humans. Some strains of E. coli benefit their hosts by producing vitamin K. They also prevent colonization of harmful bacteria in the intestine. This is representative of a mutual relationship between bacteria and humans," Jayanthi said.



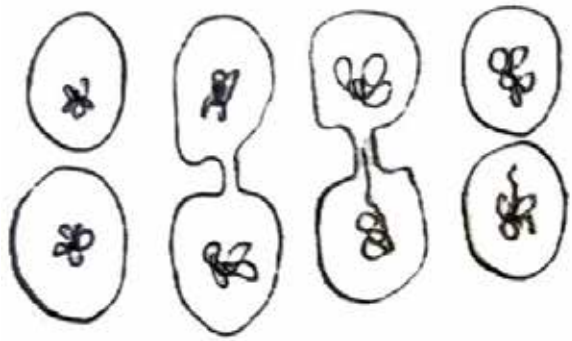
Lactobacillus E coli

Madhavi said, "But E. coli contaminated water causes stomach ache and nausea. I read it somewhere."
"Yes, some strains of E. coli are pathogenic. Open defecation is the main reason for ground and surface water pollution, including streams, rivers, lakes, and water used to irrigate crops. So, we must wash our hands with soap and water every time we use the toilet," I replied.
"Before eating food, we should wash our hands to avoid infections," Vivan added.
"Bacteria also enter through open wounds and cause diseases. An infected wound can swell, become red, feel warm, and ooze pus. Pneumonia is caused by Streptococcus pneumonia, which colonizes the nasal cavity and respiratory tract," Raja said.
"A few days back, while cycling I hurt my leg. My father washed the wound with running water. He used salt water to clean around the wound, then covered the wound with a bandage. I got cured easily," Vivan said.
"Indeed, that is the right way to treat wounds. This avoids bacterial and fungal infections," Raja said.

"How do bacteria reproduce?" Rini asked.
"Look at these pictures. Bacteria reproduce in different ways. When the cell grows to a maximum size, it divides into two identical cells. This is called binary fission. When a cell grows and divides into many cells, it is called fragmentation. When one side of the cell grows and separates, it is called budding."

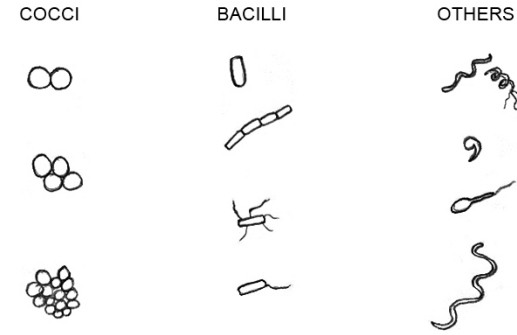


"There is another method too. Two cells come together and form a cytoplasmic bridge. The cell content flows between the two and then the cell separates. This is called conjugation," Jayanthi explained.



"I have a question. Though the place contaminated by microbes is completely cleaned, again they reappear in the next season. I noticed it on walls. How does that happen?" Anand asked.
"During unfavourable conditions, like a dry season, or when there is scarcity of food, the cells dry up and form a thick wall to protect the microbes from harsh environmental conditions. When the conditions are favourable again, these spores begin to germinate and start their normal activities," Raja replied.

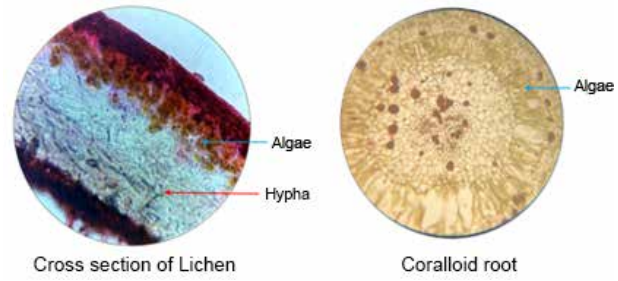
"We observed only rod-shaped bacteria. Do they exist in other shapes also?" Anand asked.
"Bacteria occur in different types and shapes in Nature. See this poster. Cocci are spherical in shape. Others may be spiral, club, or comma-shaped. In addition to this, some may have single or many flagella," Raja pointed to the poster hanging on the wall.



"Yes, we observed rod-shaped bacteria in chains," Vivan said.

Prof. Reshma entered the lab and asked us how things were going on.
Anand replied, "From the pond water sample, we observed organisms under the microscope. Paramecium was moving; we also saw some round green algae."
"Ma'am, we observed penicillium growing on lemon and hairs on fern leaves," Sujatha replied.
"Different types of bacteria also," Rini added.
"Very good. You have seen Cycas trees in the garden. Here, in the lab, you will see the cross-section of a Cycas root. The central portion of the root is called the 'pith.' Surrounding the pith is the cortex region. You can notice green algal colonies in the cortex," Reshma said.

"These are coralloid roots. I remember Nandini aunty once informed us about this," Shilpa said.
"Yes, algae help Cycas to fix atmospheric nitrogen in root cells," Reshma added.
Jayanthi called everyone to observe a cross-section of lichen and coralloid roots.



"This is great! Algae and fungi are so fragile and they made a new life together called Lichen. Is it not?" Sujatha asked.
"Yes, it's amazing. If we look at the evolution of life, algae originated earlier and Cycas later. How does algae enter the roots of Cycas and start living there? Similarly, how did the E. coli and human relationship start?" Anand asked.
"It's interesting. If you study evolutionary biology, you will learn more about the different types of relationships among various creatures in this world," I replied.
"It's a lot of information to process and we will need time," Madhavi said.
Sujatha folded her hands and said, "Reshma ma'am, thanks a lot for letting us into your department and showing the botanical garden and lab to all of us,"
"Oh! It is my pleasure. I hope the children will remember what they saw here today when they go to higher classes," Reshma replied.
"Thank you Jayanthi and Raja. I appreciate your efforts to make us understand the 'microbe' world," Madhavi thanked everyone.

Suggested activities

1. Identify different types of probiotics found in the kitchen.
2. Make a list of diseases caused by bacteria in human beings.

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- https://isappscience.org/for-scientists/resources/probiotics/
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Nandini Dholepat is a botanist who has been involved in teaching and research for more than 25 years. Currently, she is engaged as a consultant for Intellectual Property Rights for Karnataka State Council for Science and Technology, Bengaluru, and is a visiting faculty member at RIWATCH, Roing, Arunachal Pradesh. She can be reached at <nandinidholepat@gamil.com>.

Exploring the

As an early childhood educator, I have observed a profound difference between how children and adults perceive the world around them. This difference became evident in the course of doing a simple activity with my PP2 class, where children were asked to create collages using leaves. Their responses and artworks not only revealed unique perspectives but also taught me the importance of interpreting creativity through their eyes.

Activity overview

The activity required each child to collect leaves and create a collage based on some reference images I provided. These references served as a guide but left room for individual expression. As I moved around the classroom, I struggled to make sense of their creations. However, when I asked the children to describe their artwork, I was met with fascinating narratives. One child described their collage as a *pooja thali*, inspired by their family's rituals. Another said they had made a *laddu* to share with friends. A third child created a scene of a boat sailing in the sea, while another depicted a road.

Photos courtesy: Chintha Keerthana

Insights and observations

Reflecting on this, I realized that my adult mind was filtering these images through my own lens of meaning and order, while the children were communicating personal and imaginative worlds. My co-teachers also found it challenging to interpret the children's artwork, confirming that this was not unique to me.

world through children's eyes

Chintha Keerthana

I learned that children's art is less about recognizable forms and more about self-expression. This aligns with the philosophy of the Reggio Emilia approach: "The wider the range of possibilities we offer children, the more intense will be their motivations and the richer their experiences."

This experience taught me a significant lesson: children's artwork isn't about producing something recognizable; it's about self-expression and storytelling from their own worldviews.

A deeper look: abstract drawings and symbolism

A few days after the collage activity, a child began drawing random patterns and presented them to me as gifts. These drawings, though abstract, held deep meaning for the child, as I discovered later when she explained one of her paintings: red symbolized a garden, yellow represented flowers, and the white area represented a rose, which she explained was a gift for me, symbolizing her affection. Her narrative brought the abstract art to life in a way that was deeply personal to her, teaching me that children's art is often a symbolic representation of their experiences, desires, and affection.

Children's artwork is not simply a mirror of their environment but a window into their thoughts and feelings.

Why it's important for adults to respect children's perspectives

Recognizing and valuing children's perspectives allows adults to foster deeper connections, essential for meaningful growth and development. Loris Malaguzzi, the founder of the Reggio Emilia approach states that, "Children have a hundred languages, and a hundred more to think, to play, to speak... but they are robbed." This reminds us that adults must honour children's diverse ways of expressing themselves.

Furthermore, researchers Dockett and Perry (2005) emphasize that "Listening to children's perspectives promotes a shared understanding, which is key to building trust and meaningful relationships." When educators and parents appreciate children's unique viewpoints, it encourages open communication, nurturing both emotional and intellectual confidence.

This respect also promotes creativity, helping children feel safe to express themselves without the fear of judgment. By embracing children's ideas, adults model empathy, fostering a learning environment where imaginations and interpretations are valued, respected, and understood.

The importance of open interpretation

These experiences illustrate the importance of allowing children the freedom to express themselves creatively without adult expectations of concrete forms. What may look abstract or nonsensical to us holds meaning and intention for them. I realized that as educators, it's crucial to approach children's artwork with curiosity rather than judgment, asking open-ended questions and inviting them to share their stories.

Conclusion

Children's artwork provides a window into their world, their family, and their imagination. The difference in perception between adults and children isn't a gap to be closed but an opportunity to learn and appreciate their unique perspectives. As educators, embracing this difference allows us to foster a nurturing environment, where children feel valued, understood, and encouraged to express their world without the limitations of adult expectations.

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Seeing 'Missed Takes' in mistakes

Neeraja Raghavan

"There is *no room for mistakes* in my classroom!" These words from her own school teacher have remained in the mind of an experienced teacher of English (and a Head of department today) in a leading school in Hyderabad.

Accustomed to a culture that condemned mistakes and upheld the practice of *never* committing any error, this teacher (when she turned into one herself) made a few discoveries at an advanced stage of her teaching career: and it all happened through a course that she enrolled in.

Titled SEEING MISSED TAKES IN MISTAKES, she affirmed that this course helped her *get into the minds of her students*. How did this happen? She found that her students began articulating their thought processes while committing errors, as they started examining the errors that they had made. Even though her years of experience as a teacher had inevitably included correcting students' mistakes – and even discussing them occasionally – she candidly admitted that until she took this course, she had never really regarded errors as **resources:** which first required setting the ground for an honest and comfortable discussion of errors.

"It is *not* an easy task to own up to mistakes! I still remember my English teacher's dictum, after all!" She laughingly confessed. And then, she added profoundly: "If we condemn making mistakes, students will begin to *condemn learning*."

So how did this teacher go about her revelatory process? Firstly, she viewed mistakes as a very natural part of the learning process, instead of condemning them. Beginning with an honest admission of the mistakes that *she* would make when she was a student herself – and how she had often frustrated her own teachers with her repetitive patterns of errors – she found that hesitant students slowly began to open up and confess their own error patterns, with less and less shame or hesitation.

Gradually, she worked with another teacher in her English department to share her own learnings from the course. Then, she even went so far as to encourage that teacher to put into practice some of the strategies that *she had acquired* through the course.

So did this demand too much of that teacher's time? Surprisingly, not for too long, she confessed. Starting with crafting dummy worksheets, where she deliberately made the same types of errors that her students usually made, she then got her students to check these worksheets – which she claimed she had sourced from a similar grade of another school.

Naturally, as they were now sitting in the chair of a teacher, her fourth graders had great fun.

"It was a laughter riot," declared the teacher, "nonetheless, with a lot of learning!"

They began exclaiming:

"Even I make these errors, ma'am!"

"These students are not even proof-reading their work before turning it in!"

All this happened within one 40-minute period. Now that they had freely examined *another's errors*, they were more open to *revisiting their own* – with an open mind.

"Can you sit with me for five minutes, ma'am? I keep making mistakes with the use of articles…please show me how to get over this!"

This, declared the teacher, was a paradigm shift: when students began to open their minds to metacognition. "*Why* did I think thus? How can I *revisit my own thinking?*"

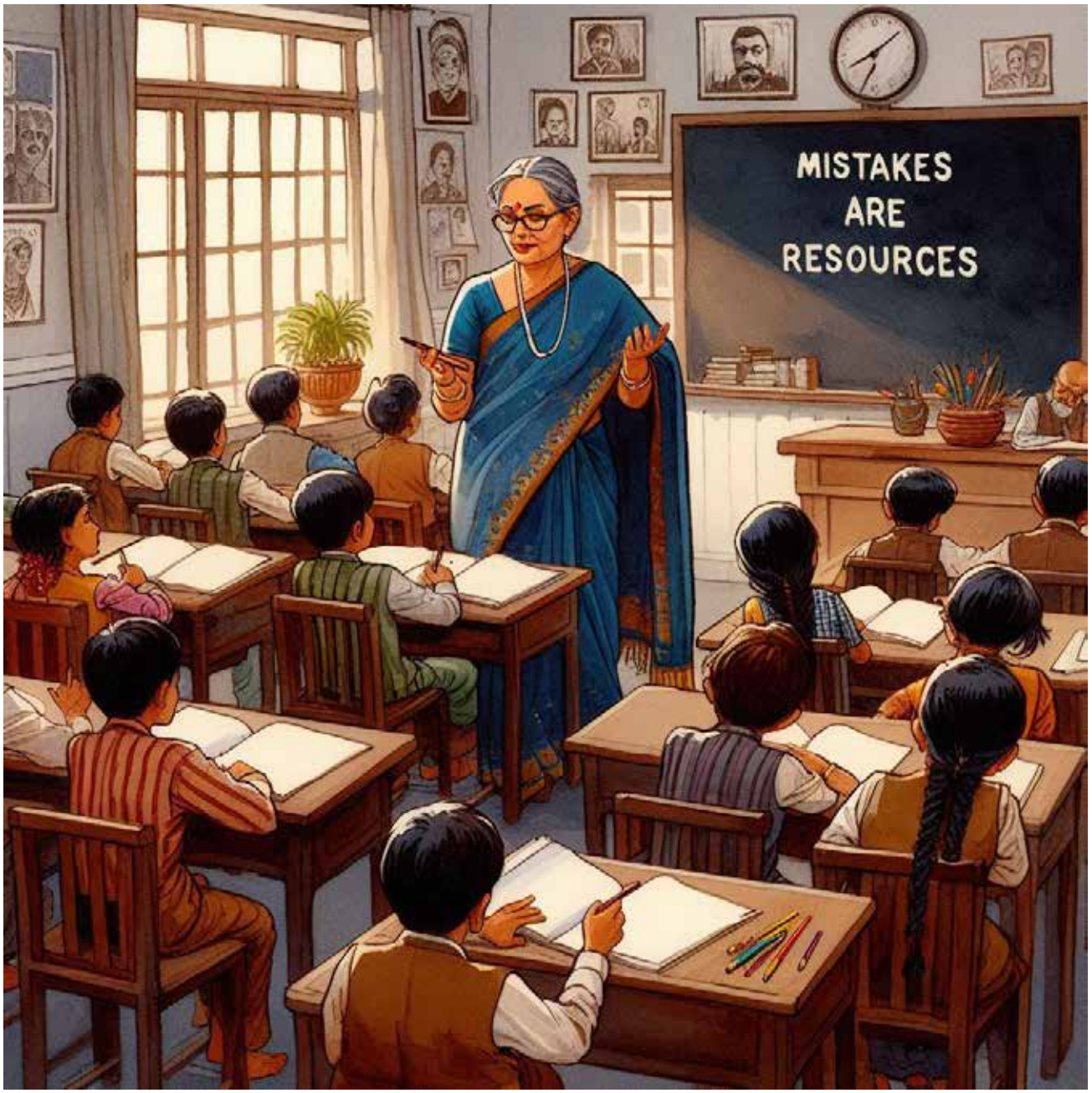


Image generated using AI. Used here for representative purposes only.

The focus then shifted from merely distributing remedial worksheets to asking why certain errors are made – and often repeatedly. Not surprisingly, student performance improved significantly within a short timeframe of five months: starting from noticeably better handwriting and timely submission of work, to actual drop in the number of repetitive errors.

She sums up her own shift thus: "Today, I see myself as a teacher-educator who welcomes mistakes in my class. Because, for me, I see the need to move beyond right and wrong and use *mistakes as a resource* to probe thought processes."

Watch her video here:
https://youtu.be/vykX9f45XpI?si=Ew4f17jX3BecABtX

Neeraja Raghavan is founder-director of Thinking Teacher (www.thinkingteacher.in), an organization that networks with teachers across the country. Thinking Teacher aims to awaken and nurture the reflective practitioner within each teacher. Thinking Teacher's goal is to help build deep inquiry and rich learning into the teaching process. The author can be reached at <neeraja@thinkingteacher.in>.

From resentment to empathy

Khrawkupar Kharshiing

Influencers. That's what they call teachers now. But how far do we go in our pursuit of being a positive influence?

Six years ago, my decision to become a teacher was not random – it was a conscious, deliberate one. I wanted to be the teacher I never had.

As a student, I wasn't one of the brightest. I was one of those kids whose name my teachers could barely remember. One of those who managed to move to the next grade without anybody even knowing I existed. I slipped through the cracks. My report card comments were the usual: "Promoted on consideration," or "Needs to work harder in the next grade," or "Weak in all subjects." The cycle continued until puberty hit.

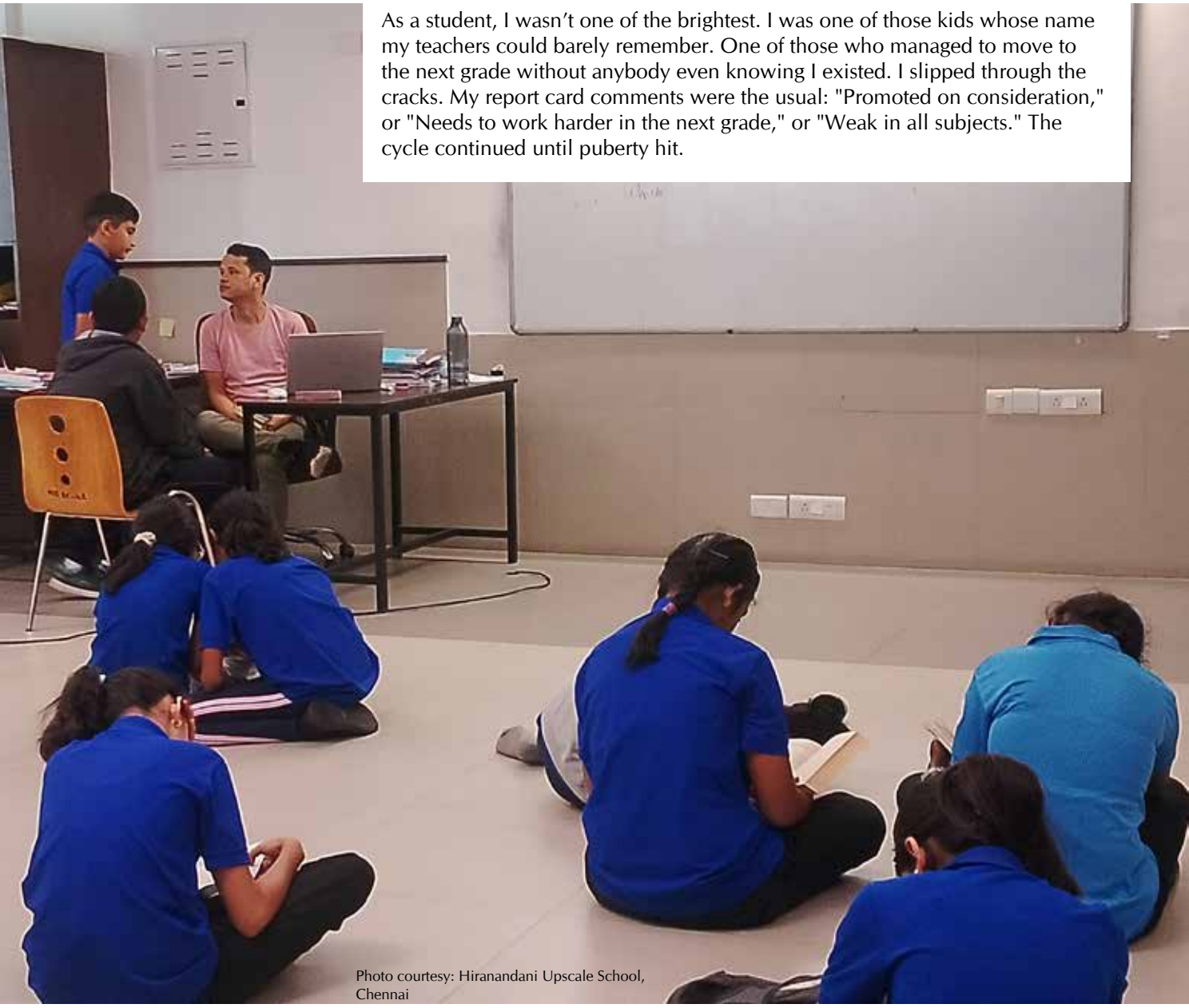


Photo courtesy: Hiranandani Upscale School, Chennai

It felt almost like divine intervention – a below-average child, hardly seen or acknowledged, suddenly started seeing blue marks on his report card. My school had a system where anyone who scored above 90 had their marks highlighted in blue. If a student failed, the marks were in red. No one understood what brought about the change. Even to this day, I'm not entirely sure what caused the change. Maybe it was a deep desire to be seen and heard, a longing to make friends, or perhaps just a shift in interest brought on by puberty.

The silent boy who avoided raising his hand in class started becoming part of the "me, me, me" group. The boy who would avoid looking at his teachers for fear of being called upon to answer a question began to ask them questions. It felt like the voice I had muffled for so long was suddenly unleashed, just like the hormones rushing through my body. My teachers never understood the change. Maybe they never tried. Maybe I never mattered to them.

But some things tend to linger, albeit with slight changes. This time, my report card had comments like "Needs to work harder in mathematics" or "Weak in mathematics." The faint blue marks on my report card were always contrasted by the glaring red ones. They all said I needed to practice! But how was I supposed to know what to practice? I tried looking at my friends' notebooks, but the equations, HCF, division, and multiplication continued to elude me.

For three years, I fought the battle – alone, mentally. My eyes tried to capture everything, but I could barely manage a score of 30. Yet, it didn't matter. They still didn't care.

I wanted to conquer mathematics. But I had no resources to help me. Nor was there anyone willing to guide me.

Then, when I was in the 9th grade, my mother, with her limited income, agreed to hire a tutor. Through a friend, I found a math tutor.

I never imagined that those 7 am classes would change my life. The tutor realized I was in desperate need of help. She pressed the reset button and decided to take it from the very beginning. I was taught subtraction, addition, multiplication, and division. Her dedication and the hope she had in me sparked something inside me – a part of my brain that had previously refused to shine. Mathematics was no longer elusive. For the first time, I scored a 54. It was a huge achievement for me. By the end of 9th grade, that blue colour finally adorned my mathematics score.

For years, I nurtured a profound respect for my math tutor. Had she not been there, I wouldn't have been able to conquer mathematics. But that respect for her also planted a resentment towards all the teachers I had had in school. 'Weak,' 'needs to study harder,' they would say. They could have done better. I hated them and always felt that my school life had been of no use. Somehow, puberty transformed me into a better student and destiny brought the math tutor into my life. My school teachers had no role in it.

However, after becoming a teacher, the resentment gradually waned. How could my teachers have focused on me when they had 50 other students in the class? How far could they have gone to be that positive influence in my life? Perhaps they were willing to help but simply lacked the energy, resources, or support to do so. Maybe that one red mark didn't matter when they had so many other students to tend to. Now, as a teacher myself, I don't harbour any hatred towards them. Instead, I empathize with the challenges they faced.

As I move forward in my journey to becoming an influencer, I've come to understand that it's perfectly okay to be like the math tutor who was willing to reset and restart when needed. However, it's also important to recognize that perfection lies in knowing what's within our control and what isn't. Our desire to influence and guide our children should never come at the cost of our mental peace, family commitments, or personal time.

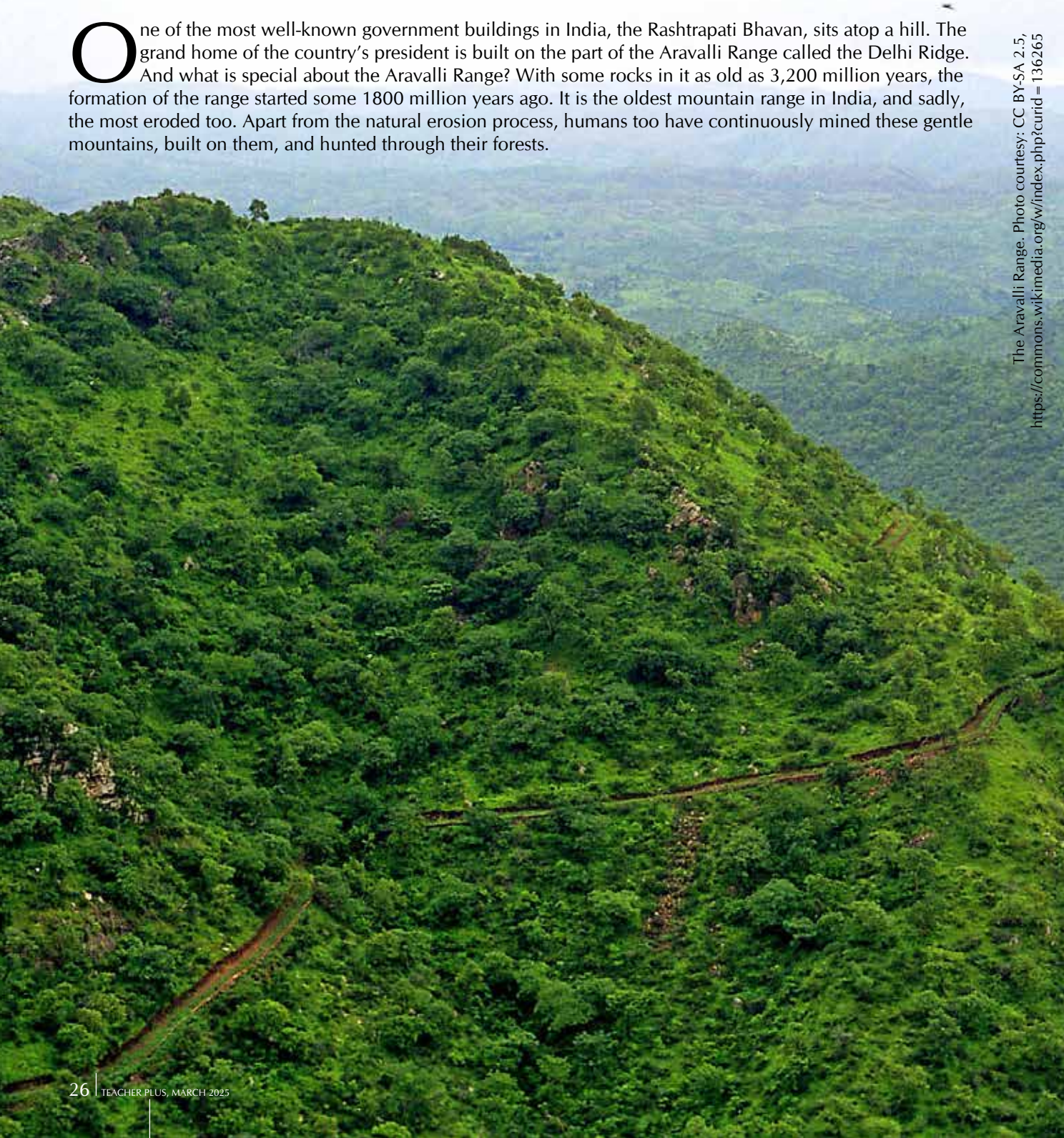
One of the school heads I worked with once said, "We are not gods. We can only do our best and leave the rest to him (god)." Perhaps, in the end, my school teachers left me to him as well.

Khrawkupar Kharshiing is a passionate learner who constantly strives to learn and apply learning in the classroom. Being in the classroom makes him happy. He finds joy in reading and expressing himself through poems when he is not teaching. His poem "Caged" was awarded the tenth rank in the National Poetry competition organized by S7. He can be reached at <kharshingkhraw@gmail.com>.

Mountains of lessons

Mala Kumar

One of the most well-known government buildings in India, the Rashtrapati Bhavan, sits atop a hill. The grand home of the country's president is built on the part of the Aravalli Range called the Delhi Ridge. And what is special about the Aravalli Range? With some rocks in it as old as 3,200 million years, the formation of the range started some 1800 million years ago. It is the oldest mountain range in India, and sadly, the most eroded too. Apart from the natural erosion process, humans too have continuously mined these gentle mountains, built on them, and hunted through their forests.



The Aravalli Range. Photo courtesy: CC BY-SA 2.5, https://commons.wikimedia.org/w/index.php?curid=136265

For most of us who do not live on the mountains or anywhere close to a hill, it is difficult to see them as living ecosystems that are integral to our lives. Behind every glass of water that we drink is the fact that it probably reached us after emerging from and flowing down a mountain. Our wellbeing depends, to a large extent, on the produce we get from the mountains, directly or indirectly. How can we as adults help our children understand the importance of mountains? An immersive excursion to a mountainous area would be the best way, but a close second would be to dedicate at least a week to learning from the mountains. Full disclosure: I am a writer, not a full-time teacher, although I have been in many classrooms as a facilitator engaging in nonformal teaching. After my book *Up the Mountains of India* was published, I've also had occasion to conduct many sessions based on the book with students in different schools, giving me some confidence in suggesting the following activities. If the school timetable allows some flexibility, maybe teachers could swap classes so that the activities can be held for two continuous periods each day, for six-to-seven days. If not, one period a day would be a good starting point too.

(This article includes a suggested list of reading via the 'Nature Writing for Children Newsletter: https://azimpremjiuniversity.edu.in/articles-nwfc/mountain-magic).

Before the week begins, students have to be assigned the task of reading as much about mountains as possible. Subject teachers need to use one period during that week for the mountain masterclass. Let's begin.

Day 1: Language class, let's say Kannada
Students carry chits of paper and pencils. Have the class sit in any place that has steps – an amphitheater outdoors or the school auditorium would be great, but a staircase would be equally effective for the experiment. The aim of the game is for students on the top step to write a line in Kannada (or any other language used in the region) about the most amazing thing they read about mountains. (Depending on the age and exposure of the children to the topic, the teacher could hint that it could be related to *anything*. It could be something like 'the Himalaya is a young mountain range', or that six of the world's staple food crops grow best on mountains – potatoes, maize, apples, tomatoes, sorghum, and barley). As the sheet is passed down, students add more interesting bits related to mountains in the language being studied and pass it down. Students on the bottom-most step read out the lines.

Like with a game of Chinese Whispers, this game may lead to some mirth and noise, and I think it would be totally worth it. Years ago, I had spent a few minutes gazing at a group of children sitting in the Panna Meena kund (also called Panna Miyah kund) in Jaipur. Were the children counting the number of sets of steps going down to the stepwell? Were they wondering at the geometry of the place? Were they amazed with the symmetry of the square structure a short distance away from the famous Amer Fort in the Aravalli mountains? All around the stepwell were stairs that for hundreds of years had served not just as a means to reach the water, but also as a place to sit and spend time with friends and family, discussing matters of importance, or engaging in the important activity of relaxing and doing nothing.

Expected outcome: Students improve their vocabulary in a particular language. They write in the language being studied. The final result is a set of short essays on the topic of mountains. If nothing, the students have relaxed for some time, making them more open to learning new things!

Mountain words: Another activity during a language class could be about mountain words. Is a hill the same as a mountain?

Till the 1970s, a mountain was a landform that rose above 1000 feet. Anything under was called a hill. But now, the two words are used interchangeably, though a hill still means something small and a mountain, big. So, what does making a mountain out of a molehill mean? Is a peak the same as a summit? Can you recall any phrases that feature a mountain-related word? How about these – peak form, mountain of dishes, uphill task, going downhill, if the mountain won't go to Mohammed, then Mohammed must come to the Mountain?

Other activities in a language class could be reading from the works of writers in the hills (Rudyard Kipling, Ruskin Bond) or reading creative work about mountains.

Eastern ghats
Photo courtesy: Vinayaraj - Own work, CC BY-SA 3.0,
https://commons.wikimedia.org/w/index.php?curid=19000353

Day 2: Social studies
Open the class to questions. What are tectonic plates? How are mountains formed? How do they affect the climate? What are the different kinds of mountains? Can volcanoes erupt from any mountain? What are the examples of wildlife in, say, the Eastern Ghats? Continue with this range as the chosen one for further activities.

This will be a theory class, where students just have to come up with questions. Alternatively, make it a quiz session with each group asking another a question. Questions get 2 points. Answers get 4. At the end of the class, there are likely to be several unanswered questions. Farm the questions to students, so that each student (or a pair of students) needs to get the answer to just one question. If a library period can be used, that will be useful. If not, this can be given as an after-school assignment.

Day 3: Science
With all the information collected as part of the social studies class, students will now probably be able to arrange it under different heads – history, mathematics, physics, chemistry, and so on. If some topics haven't come out during the fact-finding exercises, say about hot geysers in cold mountains, the teacher could introduce the topic. How do some pilgrims to Yamunotri in Uttarakhand cook their rice? They dip a bundle of raw rice into a natural hot spring! Does it take longer to cook rice in the plains or in the mountains? Why? Does the time vary depending on where the mountain is? If possible, teachers could show pictures of several mountains to showcase different topics in science – the rainbow hills in Ladakh and the marble rocks of Bhedaghat in the Satpura range (minerals and the formation of colours), Spiti's shales (fossilization of species that died maybe a 100 million years ago), the clouded leopards of Mizo Hills (long fat tails and the physics of balance), sky islands of the Western Ghats (cloud formation) and so on. A visual tour and science class rolled into one!

Day 4: Mathematics
Students of grade 6 and up probably know that Mount Everest is the highest mountain on Earth, but Mauna Kea in Hawaii is the tallest. Ask the students how they can show this in class. (Hint: Time for a student to stand up on the bench!) Talk

Bhedaghat
Photo courtesy: Aksveer - Own work, CC BY-SA 4.0,
https://commons.wikimedia.org/w/index.php?curid=75912663

Mount Everest
Photo courtesy: Luca Galuzzi, CC BY-SA 2.5,
https://commons.wikimedia.org/w/index.php?curid=1810976

about standard measurements and why we use them in various professions. In the age before satellites, drones, and GPS, how did a mathematician who never climbed a mountain calculate the exact height of Mt. Everest? Could the class calculate the height of a tree in the school compound without climbing it? Depending on the grade, there are many math activities that could be done using mountain data – ascending and descending order, conversion of units from one system to another, mountain economics (like, why are some vegetables costlier in the hills than in the plains?), what are the different shapes of mountains (sharp pyramids of the Karakoram, table mountains, dome-shaped ones, conical volcanic ones, and so on)?

Day 5: Society and ethics

The most important part of learning about mountains is to make children understand the interconnected world we live in. Already, within a person's lifetime, we are seeing the effect of deforestation and increased use of fossil fuels. Deforestation in mountains, specifically, is a major cause of landslides and floods. Along with this, there is a huge loss of endemic species. To show issues like human-wildlife conflict, construction versus destruction, and other ethical issues, it would be useful to have a role play session. Consider giving the following roles to different students: student in a hill school, farmer in a remote mountain trying to sell his vegetables in a market, a citizen in a mountain town (Ketti/Darjeeling/Mount Abu/Leh), an animal rights activist in a metro, a swamp deer or barasingha, a snow leopard, and so on. Each character makes a move, and the others have to decide whether it will affect them or not. For example, the mountain farmer asks his local government to build a road from their village across the national park to the city. Who does it affect and how? Who is right? Who is wrong? It is hoped that the students may see not just how each is connected to the other, but also how communication across stakeholders is important to find solutions.

Life skills can be a part of this class too. Some years ago, I was part of a group that interacted with children in a tiny school in a Himalayan village. We asked them to list out their problems big and small. An illustrator got them to draw what they had talked about. They also picked up one problem – too many plastic water bottles thrown away by tourists – and

Darjeeling
Photo courtesy: benjamin chia
https://www.flickr.com/photos/benneb/6619402425/

Clouded Leopard
Photo courtesy: Dr. Raju Kasambe - Own work, CC BY-SA 4.0
https://commons.wikimedia.org/w/index.php?curid=37410480

Rock Shelter, Bhimbetka
Photo courtesy: Gagnon - Own work, CC BY-SA 3.0
https://commons.wikimedia.org/w/index.php?curid=33749461

came up with ideas to upcycle them. Within a week, the children had drawn fruit-plucking tools, water filters, and irrigators using plastic bottles. The entire exercise became the content for the picture book I wrote called *How to Solve a Problem Like Himani* (Pratham Books).

Day 6: Art

Continue with the chosen range, the Eastern Ghats. The aim is to use an art form like collage to show a diagrammatic representation of a mountain range. A large pinboard would be a great way to showcase this work of group art. If a wall can be used, it would be ideal. Old newspaper works well for the pinboard or wall. Any area, at least 150 cm by 300 cm, on the floor would work too, in which case, it would be a drawing on the floor with chalk instead of making a collage.

Organize the students into groups. Share with them the idea that the drawings can be raw and representative (like the rock paintings in the Bhimbetka caves in the Vindhya Range), and need not be detailed and time-consuming (like the *araash* fresco work in Rajasthan's palaces in the Aravalli Range), and just a couple of each to start with. Group 1 makes the mountains (say, by tearing newspaper sheets to get six large triangles to represent mountains). Group 2 makes the flora (teak tree?) Group 3 makes the fauna (fishing cat!). Group 4 makes rivers. Group 5 sticks them to the mountains. If the students are using the floor or wall to draw, then the work could be done by one group after another. Placing the paper trees, rivers, animals on the drawing of the mountains would work as well as drawing them on the mountains.

Day 7: Music and theatre

Students can find songs about mountains and our natural world to sing in class. Many folk songs talk about mountains and rivers, like *Manne nambi yelelo* sung by the Irula tribe in the Western Ghats, or modern songs like *In This Together* by Ellie Golding. Wouldn't it be wonderful to adapt songs about our natural world for annual day performances? The Gillo Repertory Theatre brought out a wonderful play last year based on the book *The Ghost of the Mountains*. Poet Nirmala Putul writes in her native Santali and in Hindi. Her poem *Mountain Child* can convey many aspects of these landforms – natural beauty, the curiosity of children towards what they see, and more.

Conclusion

To wrap up, there are many things that children can do to understand the importance of mountains. Hopefully, students will read the suggested list of activities compiled towards the end of *Up the Mountains of India*. Projects are a great way for students to learn about things. All of us, and students specifically, are surrounded by too much information. I hope a project like this imparts something that is almost selfish in nature – that we need mountains for our wellbeing. That, if each of us wants a better world for ourselves, we have to let our natural resources be. We have to sustain them. Our mountains are ours – to safeguard, and to use wisely.

Mala Kumar is an independent editor and the author of *Travelling Treasures*, the *Rupaiya Paisa* series, *I want that one!*, and many other books for children. A former editor with Pratham Books, she can be reached at <mala.kumar@yahoo.co.in>.

Creating responsible citizens

Aruna Sankaranarayanan

That we are the world's largest democracy is indeed significant. Just as caring for the planet requires an ecological consciousness, a healthy democracy hinges on responsible citizenship. For many, being a good citizen involves being honest in their day-to-day lives and voting in state and national elections. But being a responsible citizen involves far more than being upright and exercising one's right to vote. For a democracy to thrive, we need an active citizenry that engages with issues that impinge directly and indirectly on our lives at local, regional, national, and global levels. And, education can play a significant role in cultivating responsible and responsive citizens.

What citizenship entails

In *Citizens*, Jon Alexander writes that citizenship is not about holding a particular passport or merely voting in elections, but a "state of engagement," where individuals see themselves as active agents capable of identifying areas that need to be reformed and taking on responsibilities to create a better future for all.

Bona-fide citizens exhibit certain characteristics, writes Alexander. They recognize that we are interdependent beings who create our own futures by actively participating in community and civic life. We deliberate the pros and cons of issues that impact

Illustrations: Soumya Menon

various networks of people in our communities and try to forge solutions that are just and fair. Regardless of our professions and varying identities, we all share a common identity – that of being a citizen, and accordingly, we need to work towards a more promising collective future.

Given the many crises that confront our world, it's not surprising that people across the globe are rather disillusioned with the state of affairs. Rising inequality, countries ravaged by war, growing levels of mental ill-health, and repercussions of climate change have many people questioning the narrative that's been instilled in us by previous generations. That you study well, work hard and be an honest citizen, and life will take care of the rest, is a fairy tale that works for a select few, typically those from privileged classes while the vast majority don't feel any forward momentum in their lives.

Alexander avers that we "need to see ourselves anew". Rather than viewing ourselves as consumers in a competitive society that pits us against each other, we need to tap into our collective humanity. Only if we reframe our belief that humans are "creative, capable, caring" people, can we get out of our current morass. Unless we exercise collaborative efforts and trust one another to work towards the common good, our future, in fact, our very survival is in jeopardy.

Alexander avers, "Citizenship is human nature," which can be 'suppressed' but never entirely extinguished. Right from prehistory, we have been practicing "campfire democracy," a term attributed to sociologist, Ronald Glassman, in "pre-colonial Africa, indigenous Australia, and among Native Americans". A 'participatory' form of government was also evident in the early civilizations of Mesopotamia and the Indus Valley. Alexander also argues that the histories of countries may be rewritten through a "citizen lens" to understand how average folk also contributed to and shaped the turn of events.

Rewriting the subject story

Though citizenship, as Alexander characterizes it, may be inherent in us, most people all over the world have been historically subjugated the world over, either by monarchs or governments, with colonized countries often bearing the brunt of oppression and exploitation. In an essay in *On Citizenship*, Romila Thapar argues that our concept of citizenship, as we currently understand it, is a modern one. In Greco-Roman times, citizenship was an "exclusive status," available to a privileged few,

today we have a more expansive and 'inclusive' view. Further, our current conception that evolved after the French and American Revolutions, presupposes the "equality of all citizens" with "rights as well as the right to assert these rights." Further, the constitution was the 'document' that would formally spell out "rights and duties of the citizen and the state towards each other."

Given that autocratic regimes are in power in many countries today, Thapar wonders whether we have 'failed' to "press home the meaning of citizenship" both for the privileged classes that continue to feel entitled and for those "who continue to think of themselves as subjects" as opposed to citizens of a state. So, educators can play a significant part by apprising students about their rights, helping them understand the historical contexts of how these rights were framed, and how they can exercise their "rights to improve society".

To be citizens, we need to have a sense of responsibility and agency. We take ownership of our problems and find ways to provide solutions while harnessing the strengths of our networks. According

to Alexander, citizenship also calls for an "expansive imagination." Rather than just accepting things as they are, we are able to "see how things could be" if we envision a different future and problem-solve how to get there. To be active citizens, we need to connect, engage, and contribute to our communities, however broadly we may define them, on an ongoing basis.

Further, the internet is a potent tool that we can use to mobilize resources and people. However, as Alexander notes, it has been "abused as a tool to undermine Citizenship." He also adds that a citizen internet has to be accessible to everyone without interference from large corporations or governments and guarantee the "right to both privacy and security." Though we may be far away from this ideal right now, it is not inconceivable to realize it if a sufficient number of people exercise their voice and rights.

The pandemic also highlighted that we are capable of putting selfish interests behind in the interest of the common good. The umpteen sacrifices that people make and the spirit of volunteerism that rises during natural disasters like the Chennai floods in 2023 shows that ordinary people are collectively capable of extraordinary feats. Education needs to highlight these aspects rather than perpetuate existing hierarchies.

The role of schools

What can schools do to nurture responsible citizens? First, we may give children more of a voice in terms of how schools are run so that they experience a sense of agency and autonomy. This includes what they study, at what pace, and how they would like to assess their own learning. A shift to project-based learning, where students identify a problem within their community, and work collectively to solve it makes education contextual, meaningful, and collaborative. With teachers serving as facilitators and mentors rather than old-school sages, students may become more active participants by planning and mapping their individual learning trajectories. Even if schools opt for a core curriculum that is common to all students, we may provide ample flexibility in terms of how students learn and express their learning.

Though most schools have a school government, these tend to mimic adult power structures with a few elites lauding over the majority, who tend to assume a passive role. Instead, schools may engage all students by adopting a more participatory approach. Students in each class, along with their teacher, may devise their own class rules rather than regulations being imposed top-down. Student leaders may be encouraged to listen to fellow students for concerns, suggestions, and feedback, creating a truly democratic space that values transparency, debate, and dialogue.

Finally, civics should not be limited to dry textbook lessons. Instead, students may be encouraged to become more civic-minded, taking on actual problems that affect their communities. The best way for students to learn about their rights as citizens is to be active and engaged ones, both within and outside school.

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THE OTHER SIDE

The extracurricular kid

Nurturing talent beyond

Anuradha C

The anxiety of parents and the struggle of schools

Every parent dreams of their child excelling in school – good grades, appreciation from teachers, and a promising future. But what happens when the child struggles with academics and instead shines in a sport, music, painting, dance, or some other skill? Many parents experience anxiety, fearing that their child's future might be at risk due to poor grades. Meanwhile, schools, bound by rigid curricula, often fail to accommodate and nurture these children in a way that truly recognizes their strengths. The result? A cycle of self-doubt and frustration for the child and parents alike.

Photos: Sakti Prasanna Mohanty
Courtesy: DAV Public School, Pokhariput

academics

The Indian education system, for the most part, still prioritizes textbook knowledge over talent-based learning. While some progressive institutions and alternative schools support artistic and sports-oriented children, mainstream education often falls short in fostering and honing special skills while ensuring that children are not made to feel like failures for struggling in academics.

So, how do parents and educators deal with this dilemma? How can a child's natural abilities be encouraged while ensuring that essential academic knowledge isn't completely ignored? And most importantly, when and how should a child consider making their talent a career choice?

Identifying the special skill

Recognizing a child's talent early is crucial. This doesn't necessarily mean waiting for them to win a medal or a competition but observing where they naturally excel.

1. **Interest and enjoyment:** A child who is deeply engrossed in sketching for hours, tapping rhythms on tables, or spending every free moment on a football field is displaying signs of passion.
2. **Effortless skill:** Some children grasp musical notes, dance moves, or sports techniques faster than others, often without formal training.
3. **Recognition from others:** Teachers, coaches, or even family members may notice and acknowledge the child's unique abilities.

Parents should engage their child in various activities early on and observe what excites them most. Allowing the child to explore activities that interest them, without forcing them into pre-determined career paths can lead to genuine talent discovery.

Methods to hone the skill

Once a child's special skill is identified, nurturing it is the next step. This requires structured efforts from parents, educators, and mentors.

1. **Formal training:** Enrol the child in professional coaching, workshops, or specialized classes. A good mentor can guide skill development in a structured manner.
2. **Practice and exposure:** The 10,000-hour rule suggests that mastery comes with consistent practice. Encourage daily efforts and provide exposure through competitions, exhibitions, or performances.
3. **Financial and emotional support:** Specialized training, equipment, and travel for competitions can be expensive. More than financial investment, emotional support and encouragement from parents are vital for confidence-building.
4. **Balancing pressure:** While discipline is important, excessive pressure can backfire. Encourage progress, but don't turn passion into an obligation.

Marrying basic academics with skill development

While academic excellence may not be the strong suit of an extracurricular-driven child, basic literacy and numeracy are essential in today's world. The challenge is to create a balance where academics do not overshadow talent but still remain a part of learning.

1. **Customized learning plans:** Schools can offer flexible academic programs where core subjects (math, language, and science) are taught in ways suited to different learning styles.

2. **Skill-based curriculum:** Alternative schools like Rishi Valley and The Valley School blend skill development with basic education, allowing children to thrive in both areas.
3. **Technology as a tool:** Online learning platforms allow children to study at their own pace, reducing the pressure of traditional schooling while ensuring academic progress.
4. **Open communication:** Parents and schools must collaborate to create a supportive ecosystem, where the child doesn't feel isolated for performing poorly in traditional academics.

When to make the special skill a career choice

This is one of the biggest dilemmas for both children and parents. Making an unconventional skill a full-time career is daunting, especially given societal norms that prioritize traditional professions like engineering, medicine, or law. Here's when a talent can be considered a viable career:

1. **Consistent excellence:** If a child consistently outperforms others in their skill area and shows a rapid learning curve, it indicates career potential.
2. **Recognition and opportunity:** Winning state or national-level competitions, being scouted for professional training, or receiving opportunities in their field are strong signs.
3. **Sustainable earning possibility:** Researching income opportunities in the field and understanding financial sustainability is crucial. With platforms like YouTube, online teaching, freelancing, and sports scholarships, monetization of talent is more accessible than before.
4. **Passion over pressure:** If the child sees their talent not just as a hobby but as something they genuinely want to dedicate their life to, it's worth pursuing.

Indian examples of special skill careers

1. **P.T. Usha (Athletics):** Born in a modest family, she pursued running despite the lack of sports infrastructure in her village. Today, she is one of India's most celebrated athletes.
2. **Lydian Nadhaswaram (Music):** A young piano prodigy from Chennai, he won the international reality show "The World's Best" at the age of 13, bringing global recognition to Indian musical talent.
3. **Biswa Kalyan Rath (Stand-up comedy):** An IIT graduate, he pursued comedy despite societal expectations of a conventional job, proving that creativity can be a career.
4. **Rameshbabu Praggnanandhaa (Chess):** At just 16, he has defeated world champions and shown that chess, often seen as an extra activity, can be a career path.
5. **Sudarsan Pattnaik (Sand art):** Despite being from a poor background, his perseverance turned sand art into a globally recognized skill, earning him multiple awards.

Conclusion

Not every child is meant to be a doctor or an engineer, and that's okay. The world needs athletes, artists, musicians, and innovators just as much. The responsibility of parents and educators is to nurture these talents without forcing them into pre-defined moulds.

While academics are important, they should not be the only measure of intelligence or future success. With the right support system, customized learning approaches, and societal acceptance, extracurricular-driven children can achieve greatness in their own fields – without being labelled as academic failures.

Encouraging passion, fostering talent, and making informed career choices will ensure that every child finds success in a field where they truly belong.

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Implementing global citizenship education in India

Challenges and considerations

Dr. Gh Hassan Wani

In a world grappling with complex global challenges – ranging from climate change and social inequality to cultural diversity and digital connectivity – Global Citizenship Education (GCE) has emerged as a transformative approach aimed at fostering global understanding, tolerance, and responsibility. GCE, as defined by UNESCO, promotes civic engagement, cultural awareness, and an appreciation of global interconnectedness. For a country as diverse and populous as India, implementing GCE could be transformative, offering young people the tools and perspectives necessary to navigate and contribute to an interconnected world. However, the path to successfully integrating GCE in the Indian education system is fraught with challenges, including cultural diversity, socio-economic disparities, limited resources, and a traditionally structured curriculum. This article explores the unique challenges and critical considerations in implementing GCE within the Indian educational context.

Defining Global Citizenship Education in the Indian context

A primary challenge to implementing GCE in India is defining and contextualizing it within India's diverse socio-cultural landscape. GCE is a relatively new concept in Indian education, and its goals – such as fostering a sense of global belonging, empathy, and respect for diversity – may be seen as either abstract or secondary to traditional academic goals. Moreover, India's national identity, rooted in pluralism and secularism, intersects in complex ways with the universalist ideals of GCE.

To address this challenge, Indian education policymakers, and curriculum developers need to adapt GCE goals to local values and contexts. India's pluralistic heritage and commitment to secularism can form a strong foundation for GCE, making it easier to integrate global and civic education elements. Emphasizing the compatibility between GCE's goals and India's own values of tolerance and inclusivity could make GCE more relevant and relatable to students, educators, and parents.

Cultural and social resistance

India is a country of deep cultural traditions, where regional, religious, and linguistic identities are closely guarded and valued. GCE, which encourages global empathy and often addresses issues such as human rights, gender equality, and cultural diversity, may face resistance from certain segments of society. Some might view GCE as a threat to local or national identities, believing that it could dilute or undermine traditional values. For example, discussions on gender roles, environmental justice, and human rights, central to GCE, may clash with local beliefs or practices.

To overcome this challenge, educators must frame GCE in a way that respects and incorporates India's cultural values, highlighting how a global perspective can coexist with, and even strengthen, local traditions. An effective approach could involve emphasizing "glocalization" – the idea that students can be active participants in their local communities while also embracing a sense of global responsibility. By highlighting that local identities are not at odds with global citizenship, Indian educators can help alleviate cultural resistance and foster a more inclusive approach to GCE.

Language and accessibility

India's linguistic diversity presents a significant challenge to GCE implementation. With 22 official languages and hundreds of dialects spoken across the country, creating standardized GCE resources that are accessible to all is challenging. Many GCE materials and resources available globally are primarily in English, which is inaccessible to a large portion of India's rural population.

One solution is to develop multilingual GCE resources, allowing students from different linguistic backgrounds to access and engage with GCE content. Translating GCE materials into major regional languages and incorporating culturally relevant examples can help make global citizenship concepts more relatable and understandable. Partnerships with local NGOs, translation agencies, and community

leaders can facilitate this process, ensuring that GCE content is accessible to diverse linguistic groups.

Socio-economic disparities and inequitable resource distribution

The Indian education system is marked by stark socio-economic disparities. While urban schools, especially private institutions, have access to better resources, technology, and teacher training, many rural and under-resourced schools struggle to meet basic educational needs. Implementing GCE, which often involves interactive teaching methods, digital resources, and experiential learning, may be particularly challenging in resource-limited settings.

To address this challenge, GCE implementation in India may need to be adapted to leverage existing resources in innovative ways. For instance, experiential learning activities – such as community engagement projects or local awareness campaigns – do not require significant resources and can be effective tools for imparting global citizenship skills. Additionally, mobile technology and digital platforms can bridge the gap, with low-cost or government-sponsored initiatives providing digital access to GCE content for under-resourced schools. However, government and community investment in equalizing educational opportunities is crucial to ensuring that all students benefit from GCE.

Curriculum overload and rigid academic structure

Curricula across India's school boards are often heavily structured and academically rigorous, leaving little room for new subjects or non-traditional topics like GCE. The pressure of standardized exams, competitive college admissions, and an exam-oriented culture means that schools and students may be reluctant to adopt new educational programs, especially those that are not immediately linked to academic achievement or career advancement.

To integrate GCE effectively, it may need to be woven into existing subjects rather than introduced as a standalone course. For example, environmental science classes can discuss climate change from a global perspective, history classes can include lessons on global migration patterns, and civics classes can explore international organizations and global issues. By embedding GCE content within established subjects, educators can circumvent the challenge of curriculum overload and gradually introduce students to the ideas of global citizenship.

Teacher training and preparedness

Teachers play a central role in implementing GCE, yet many Indian educators lack the training and support needed to teach global citizenship topics effectively. GCE requires a unique skill set that includes knowledge of global issues, cultural sensitivity,

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critical thinking, and interdisciplinary teaching methods. Unfortunately, teacher education programs in India have traditionally focused on subject-specific knowledge, with limited emphasis on GCE themes or pedagogical strategies for interactive learning.

To address this gap, professional development programs for teachers are essential. Workshops, seminars, and online courses can help educators develop the skills needed for GCE. Government support for teacher training in GCE, along with collaboration with NGOs and international organizations, can further enhance the reach and impact of these programs. Building a network of educators who are trained in GCE can also create a supportive community for sharing ideas, resources, and best practices.

Political sensitivities and regional diversity

India is a politically diverse country with a complex socio-political landscape. Certain topics central to GCE – such as human rights, environmental justice, and migration – can be politically sensitive and may be interpreted differently in various parts of India. For instance, discussions about migration might be received differently in regions with high migrant populations compared to regions that are culturally homogenous.

Implementing GCE in India thus requires a nuanced approach that takes into account regional diversity and political sensitivities. Teachers and curriculum developers should be encouraged to present global issues from a balanced perspective, allowing students to form their own informed opinions rather than pushing a particular political stance. Emphasizing critical thinking and open discussion can help students explore these issues constructively while respecting different viewpoints.

Lack of standardized assessment for GCE competencies

Assessing GCE competencies presents a unique challenge, as traditional assessment methods are ill-suited to evaluating values, attitudes, and skills such as empathy, critical thinking, and civic responsibility. In India, where assessments are highly exam-oriented and focused on academic performance, introducing new assessment methods for GCE can be challenging.

Innovative assessment approaches, such as project-based evaluations, self-reflection exercises, and peer assessments, can provide a more holistic view of a student's progress in GCE. For instance, students could be asked to work on projects that address local issues from a global perspective, allowing them to apply GCE concepts in a real-world context. While these alternative methods may be challenging to scale, they provide valuable insights into students' understanding and personal growth, beyond academic achievements.

Encouraging local engagement and community-based learning

An essential aspect of GCE is empowering students to engage with and act on local issues with a global perspective. However, India's large, diverse population, and varying socio-economic conditions mean that community engagement efforts must be tailored to local needs and conditions. In many Indian communities, issues like clean water access, sanitation, and waste management are more immediate concerns than global topics like climate change or international migration.

To foster meaningful engagement, GCE in India should emphasize local action that has global relevance. For instance, environmental conservation projects or health awareness campaigns allow students to address issues that are both locally significant and globally relevant. In this way, GCE can build a bridge between students' immediate surroundings and the wider world, making global citizenship concepts tangible and actionable.

Conclusion

Implementing Global Citizenship Education in India holds immense potential to create a generation of young people equipped to navigate and address global challenges with empathy, critical thinking, and a sense of responsibility. However, challenges ranging from socio-economic disparities and curriculum rigidity to cultural diversity and political sensitivities must be thoughtfully addressed.

By contextualizing GCE in a way that respects India's cultural values, investing in teacher training, leveraging technology, and focusing on local-global connections, India can make GCE accessible and relevant to students across its diverse landscapes. As Indian policymakers, educators, and communities embrace and address these challenges, GCE can become a vital tool in shaping a more inclusive, empathetic, and globally connected society.

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Getting ready to take on the world

Monica Kochar

In today's interconnected world, classrooms cannot operate in isolation from global trends and challenges. The rapid advancements in technology and increased movement of people across borders have created an environment where local and global realities are deeply intertwined. Schools now reflect this diversity more than ever with students from various racial, cultural, and linguistic backgrounds. This demands that teachers adapt strategies to meet these evolving needs and prepare students to navigate the complexities of a globalized world.

Integrating global competence: a three-step approach

To embed global competence into teaching practices, I recommend a simple three-step approach:

Step 1: Define the skills

Clearly identify the competencies students need to thrive in a globalized world. These might include critical thinking, cultural awareness, empathy, and problem-solving. By focusing on specific skills, teachers can create targeted learning opportunities.

Example: A math teacher could focus on developing students' data interpretation skills while fostering their understanding of global health trends. For instance, students could examine data on vaccination rates across countries, analyze the factors influencing these rates, and discuss the implications for public health.

Step 2: Create engaging tasks

Design tasks that actively engage students in practicing these skills. Effective tasks connect academic concepts to real-world challenges, encouraging students to think critically and collaborate.

Example: Using geometry, students could explore urban planning by designing a sustainable city layout. They could calculate areas for green spaces, analyze traffic flow patterns, and consider environmental factors like reducing carbon footprints. This task not only reinforces mathematical concepts but also introduces students to sustainability and global urban challenges.

Step 3: Develop an assessment rubric

Assessment is crucial for measuring progress and ensuring that students internalize the skills and values they learn. A well-rounded assessment includes teacher evaluation, self-assessment, peer feedback, and reflections on real-world relevance.

Example: For a project on renewable energy, students could calculate the efficiency of solar panels using trigonometry. The assessment rubric could evaluate the accuracy of their calculations, their ability to present findings, and their reflections on how renewable energy can address global energy needs.

Real-world applications in math

Mathematics provides a powerful platform to connect students to global issues. Here are two examples.

Understanding climate change through graphs

Climate change is a pressing global challenge that can be explored through mathematical concepts like graphs. Students analyze real-world data, such as trends in global temperatures or carbon emissions to understand the mathematical relationships behind these issues.

Task: Students plot temperature changes over the last century on graphs, focusing on a specific region. They analyze the data to identify the trends, discuss contributing factors like deforestation or industrialization, and propose actionable solutions to mitigate climate change.

Impact: This activity not only reinforces graph skills but also fosters critical thinking and emotional

engagement as students realize the tangible effects of environmental changes on communities worldwide.

Exploring global health through probability
Probability can be used to understand global health challenges, such as the spread of diseases. By examining real-world data, students can learn how mathematical models predict outcomes and inform policy decisions.

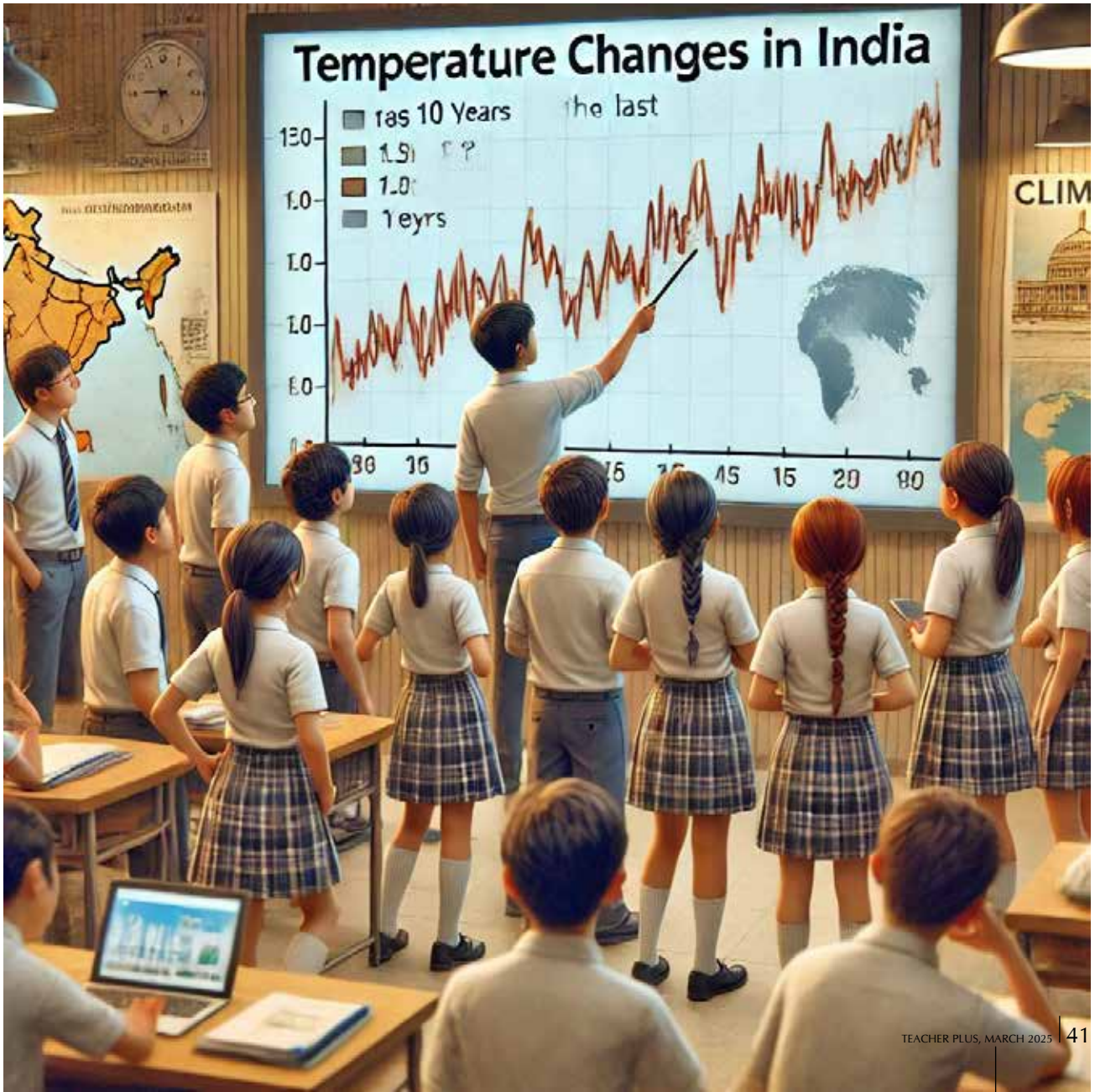
Task: Students analyze data on the probability of disease outbreaks in different regions. They create probability models to predict the spread of a disease based on factors like vaccination rates or population density. Students then discuss how mathematical predictions can guide public health strategies.

Impact: This task enhances students' analytical skills and deepens their understanding of how mathematical tools contribute to solving global health challenges.

Assessment: Measuring growth in global competence

Assessment plays a pivotal role in fostering global competence. It ensures students engage deeply with the material and develop the necessary skills and values. A comprehensive assessment approach includes:

Teacher assessment: Evaluate the accuracy and clarity of students' mathematical work, such as their graphs or calculations.



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Self-assessment: Encourage students to reflect on their learning process and how the task influenced their understanding of global issues.
Peer assessment: Facilitate collaboration by having students review each other's work and provide constructive feedback.
Reflection: Ask students to write about how the task changed their perspective on global challenges and their role in addressing them.
Example: In a task exploring water scarcity, students could calculate their daily water usage and compare it with global averages. The assessment could include evaluating their calculations, their ability to present solutions for reducing water waste, and their reflections on the importance of water conservation.

Emotional engagement and real-world relevance
To truly foster global competence, lessons must evoke an emotional response. When students analyze climate data, they may feel a sense of urgency about environmental sustainability. Similarly, examining global health data can spark empathy for affected communities. These emotional connections make learning meaningful and inspire students to take action.

Example: After studying renewable energy, students could design a campaign to promote solar energy in their communities. This project not only reinforces their learning but also empowers them to make a tangible difference.

Conclusion

Fostering global competence is not just about preparing students for academic success, it's about equipping them to contribute meaningfully to a globalized society. By embedding global competence into teaching, we help students become informed, responsible citizens, ready to address the challenges of an interconnected world.

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Monica Kochar is a seasoned educator and curriculum designer specializing in math education. With over 10 years of experience, she created HumaneMath, a human-centric math program, and has worked with leading organizations in edtech, publishing, and training. Her expertise spans curriculum design, teacher training, assessments, and quality review. Monica has contributed to key startups, including WhiteHat Jr. and mentored several initiatives. She holds a Master's in Education from the University of the People. She can be reached at <humanemath@gmail.com>.

Cultivating

For the last five years a small team of the Fig Tree Learning Centre has been working in three government primary schools north of Bengaluru. We regularly visit the schools and work with the children on language and environmental studies.

Last year, we invited the 3rd and 4th standard children of the schools to suggest a theme for the display of books, visuals, and their own writing in the classroom. Earlier we had done displays on friendship, travel, insects, wild flowers, etc., and we assumed a similar 'doable' topic would come up. Imagine our surprise when the children eagerly declared that the display should be about gods and ghosts! This is a subject that is rarely included in the textbook, and yet it was clearly a significant part of the children's lives. Further, having offered them a choice, some response from us as facilitators was demanded.

In addition to our team sharing a number of carefully chosen books (that included humour and legends) on the subject of ghosts, the children also contributed their artwork. Some wrote about their own or of others' encounters with ghosts. The children had no difficulty in visualizing ghosts. Some wrote in detail about their habits, the sounds, the appearance, and antics and powers of ghosts. Also, the children took turns to be in 'the hot seat' to respond to questions from classmates about the ghosts they had so readily conjured up. In expressing their feelings and experiences, there was perhaps a process of naming and giving these often menacing characters a shape, colour, sound, and action to what had remained unsaid. In the act of sharing, there was the beginning of a sense of control and confidence that this too could be faced.

As librarians, we felt that we could not leave the subject there. It was true that in doing these

hope in the classroom

Jane Sahi

activities, there was a certain light-heartedness and humour, but there was also an underlying sense of children's vulnerability and anxiety. The question we asked ourselves was how we could share about good and positive forces without being sentimental or dishonest. On the suggestion of a friend, we turned to the book, *A Kite Called Korika*[1] to help children think about their own dreams and aspirations. In the story, a nine year old Yellaiah finds a kite and names it Korika, which in Telugu means 'wish/desire.' Yellaiah and his friends discuss their wishes – bangles for one, a full meal for another, and a schoolbag for a third. Yellaiah himself dreams of having a cycle of his own. Meanwhile, Yellaiah's younger brother, Mallaiah, is unwell and is so troubled by his fear of death that he refuses to go out or play with his friends. Yellaiah finally foregoes his wish and along with his brother, sends a message with his kite that his brother Mallaiah may overcome his fears and play happily once more. The story of Korika does not foster illusions that something magically will resolve problems, but subtly opens an understanding of what wishes are about and that we are supported and guided by positive forces in ways we cannot always see or touch.

Then the children in the class made their own kites and wrote their wishes on them. We were surprised by their responses. Apart from one child hoping for a cycle, the other children's wishes were often for solutions to family problems – a loan that needed to be repaid, a grandmother's recovery from sickness, for father to find work, or a sister's baby to learn to walk. There were also personal concerns, particularly about failure in school, one girl wished she could 'get *buddhi*', another that she should learn to read well, and yet another child hoped he would pass the upcoming exams.

1. *A Kite Called Korika* by Sharada Kolluru, illustrated by K.P. Muraleedharan, Tulika Books, 2012.

The classroom can provide a safe space for children to share their fears and dreams, if they are ready to do so, through talking, artwork, role play, or writing. In writing a diary, children might share about the delight of getting a new pair of shoes, the disappointment of not being able to join a dance because the family could not afford the costume, the excitement of playing hide and seek, or the sorrow of losing a friend, the wonder of the pattern of new life, decay and fresh beginnings in Nature, or the flash of understanding something hitherto unknown. Often, these feelings come out vicariously. I remember one child with visual impairment writing and illustrating a story about the frustration of a bull until the angry creature was finally given glasses and could see the world.

The natural resilience of a child is something to be respected and nurtured. The heroes that are often presented to children are those who have defeated

Making kites at Tarabanahalli

and triumphed through strength and physical prowess, but there is another kind of hero who does not lose hope and retains a sense of purpose even in the face of loss. The Polish poet, Anna Swirszczynska recalls her father's tenacity and courage when his lifetime's artwork is destroyed by bombing. The poem is entitled "He did not jump down from the third floor" and it ends with the lines:

His workshop had
no roof
no walls
no floor.

My father did not jump down
from the third floor.
My father just began all over again.[2]

All children hope in some way to be engaged, to succeed, and to feel confident in what they are doing. No child wants to fail or look foolish in front of their peers, but hopefully teachers can value hesitancy, mistakes, and tentative explorations, so that failure is not a final judgement but an accepted part of real learning. Sometimes, children give up trying to succeed when they are scarred by a negative experience where they, or their families, are blamed for their inability to learn to read or grasp a concept. Such children may remember being told that they are lazy or stubborn or just 'dumb'. Older children who have difficulties with reading often have memories of feeling powerless or bewildered that, though they tried, they could not make meaning for themselves nor could they meet the expectations of teachers or parents. The result is that too often children give up or withdraw as far as possible from any engagement.

Displaying different children's reading achievements on the wall for all to see may not serve as an incentive for struggling readers to try harder. On the contrary, it may cause some children to feel demotivated and give up. The classroom needs to be a place not of comparisons, but of inclusive practices. Within the classroom could children be given differentiated instruction to meet diverse needs, paces of working, and styles of learning? Can we look at ways of collaboration that allow children to contribute positively to a joint outcome, such as the creation of a wall newspaper, or following through an experiment together, or creating a drama?

2. He did not jump down from the third floor' by Anna Swirszczynska, translated by Susan Bassnett and Piotr Kuhiwczak in *A World of Poetry* Edited by Michael Rosen, Kingfisher, 2003.

As teachers or facilitators, we are not in a position to intervene directly to resolve problems of such enormity as the disastrous effects of a delayed monsoon or the flooding that happens in part because of misguided agricultural practices of monocropping or deforestation. We cannot easily solve the problems of inequity, which make some communities so vulnerable that they have little access to healthcare or even adequate shelter and nourishment. Yet, teachers can be a part of imagining and trusting that each one can contribute to make change possible by deepening an awareness of the root causes of injustice or exploitation.

There are practical ways in the home, school, and community for children to be encouraged to care for one another and act responsibly towards natural resources. Children also learn through a greater understanding of facts about our changing world. But we also need to think how children can imagine a more just and harmonious world. What, for example, are the stories that could help us all live creatively, peacefully, and responsibly?

I heard of a community centre in one of the most deprived and disturbed areas of Melbourne, which became a no-go area even for the police. The facilitators encouraged the youth of that place to

Sharing a ghost

make a frieze that stretched across the street. On one side, young people painted what the actual situation of that community was – the violence and danger with gangs and bullying, sale of illicit liquor and drugs, and on the other, they painted pictures of how they imagined it could be different. Imagining a different kind of future can begin to make change possible.

Planting a sapling in the school garden.

The question is how do we prepare children for an uncertain future without overwhelming them with a sense of despair and helplessness? How can we help children look to the future with hope, joy, and energy? How can the crises of our times become opportunities to re-think how we can share our resources?

The reality of a toxic world marred by pollution, violence, displacement, and loss is all too apparent to children. For many, this exposure to violence is sadly grounded in lived experience, but for others it is gleaned virtually from overheard adult conversations, TV, and the media. We cannot protect children from this reality anymore than King Śuddhodana could hide death, pain, decay, weakness, and sorrow from his son, Siddhartha (Gautama Buddha). Hope is not always visible; but false optimism is not helpful. Children's trust in the world is closely related to the way adults respond to the confusions, challenges, and contradictions that are inevitably part of life.

There is a place for the happy ending, where the protagonist after enduring all kinds of terrors, trials, and tribulations is finally victorious; a time and place where dark forces are destroyed and good finally triumphs. However, we all, whether as adults or children as they mature, struggle to trust in a bigger picture and a longer sense of history that brings together the past, the present, and the future. In the book, *Postcard from the Lushai Brigade* by Hannah Lalhlanpuii (Penguin Random House), the author describes how the young boy's dreams and hopes are crushed when he realizes that Au, his beloved older brother is dead. He has to face the pain of separation knowing that his brother will not return from an unjust war fought only for the benefit of the colonial powers who occupy his homeland of Mizoram.

The young boy is drawn to the place where he and his brother had shared so much and where his community believed that village heroes are "allowed to rest". He says, "I turned my head towards the forest and listened more closely to the songs of the birds and cicadas. My brother's spirit was in there somewhere, among the trees and the many sounds of nature ... I felt the warmth of another tear ... Au, I will come looking for you, every evening before sundown among the trees where the birds and cicadas sing..."

There are so many ways in the classroom and community that teachers can help children to come to know and cherish the world and to feel confident that each person can contribute something to sustain this fragile Earth.

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Dear teacher, I have so much to say

Bhuvaneswari B and Sapna Bhat

Autism spectrum disorder (ASD) and attention deficit hyperactivity disorder (ADHD) are neurodevelopmental conditions accompanied by differences in brain development. ADHD is characterized by difficulties with attention and/or hyperactivity and impulsivity. ASD as defined by Diagnostic and Statistical Manual 5* is characterized by enduring and impairing difficulties in social communication/interaction and restrictive and repetitive behaviours, interests, and activities, or sensory abnormalities. These two conditions usually co-occur. Mainstream schools can be difficult for such children for a variety of reasons. The following is a fictional description of a child who has been diagnosed with these disorders and what he might tell his teachers to help him succeed in school.

D is a curious 11 year-old boy with autism and ADHD in a general education classroom. He is the single child of very accomplished parents who care deeply about his wellbeing. D loves to read anything he gets his hands on, however he particularly enjoys reading about science, history, and computers. He loves coding, board games, and art. He finds physical activities difficult and is very aware of the differences in his abilities in comparison to those of his peers. He is above grade level on reading, maths, and writing. However, it is sometimes difficult for him to work on these subjects in the classroom as he finds it distracting and occasionally overwhelming. D thinks social situations and friendships are puzzling and in his own words he thinks he has “dyslexia for social situations”. He is aware of his ADHD diagnosis but doesn’t know that he has autism. D has a hard time with self regulation and being able to use learnt strategies when his emotions get out of control, such as raising his hand, and being able to keep his thoughts in his head. He also has intrusive thoughts about peers, which he finds difficult to control including thoughts like “friends ganging up”, “peers spying”, etc., and this interferes with his functioning in the classroom and on the playground. D’s brilliant mind and creativity remain hidden under the label of bad behaviour and perceived rudeness and disrespect. His peers avoid being paired with him in group activities and his teachers find it difficult to deal with his dysregulation when teaching a big class.

D loves school and his teachers, and wants to make friends. With the new school year starting, he wanted a change in his situation at school. At the behest of his parents, special education teacher, and speech language therapist, he set out to write this letter to help his teachers understand him.

DEAR TEACHER, I HAVE SO MUCH TO SAY...
I MAY NOT LOOK AT YOU WHILE YOU SPEAK.
BUT I AM LISTENING.

Eye contact can be overwhelming for me, and it takes a lot of energy to process everything happening around me. Even when I'm not looking at you, I am paying attention to your words.

I MAY NOT ALWAYS USE SPEECH TO TALK TO YOU... BUT I HAVE SO MUCH TO SAY.

My thoughts and feelings are just as rich and complex as anyone else's. I use a device (computer or tablet) sometimes to share my ideas and needs. Your patience and encouragement mean a lot to me.

I SOMETIMES TAKE A LONG TIME TO ANSWER...
BUT I AM TRYING.

Formulating responses and using my communication devices can take extra time. Please wait for me and give me the space I need to express myself. You can say something like, “D, it will be your turn after B’s and C’s.”

*The Diagnostic and Statistical Manual of Mental Disorders (DSM-5) is used by clinicians and researchers to diagnose and classify mental disorders.

I FEEL VERY ANXIOUS WHEN THE ROUTINE IS UNPREDICTABLE. BUT I WANT TO PARTICIPATE.

Changes in routine can make me feel lost and anxious. A consistent schedule helps me feel safe and ready to engage in activities. When things change, it helps if you can prepare me in advance.

I LIKE TO USE PICTURES TO UNDERSTAND AND COMMUNICATE.

Visual supports like pictures and written words help me make sense of instructions and concepts. They also help me express my thoughts and feelings more clearly.

I DON'T LIKE MELTDOWNS EITHER, BUT MY BODY AND MIND DON'T ALWAYS WORK TOGETHER TO LET ME EXPRESS WHAT I MEAN.

When I have a meltdown, it's often because I'm overwhelmed or can't communicate my needs effectively. Your understanding and calm during these times help me feel better. Please give me a safe space to calm down, don't ask me too many questions or ask me to problem solve at this point, it can make me feel more overwhelmed.

I WANT TO LEARN, I WANT TO LISTEN.

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I am excited to be in your class and to learn new things. Your support and understanding help me participate and succeed.

I LIKE WORKING WITH OTHERS, SO HELP ME BY SETTING THE GROUP RULES.

Groups get disorganized for me sometimes, please choose my group-mates carefully. Setting some ground rules for the group will help us function well together.

SENSORY STIMULUS LIKE NOISE AND BRIGHT LIGHTS CAN OVERWHELM ME.

Sometimes I might need noise blockers, chewy toys, or a quiet space to function well. Initially, I might need you to offer me these choices and later on I might gradually begin advocating for my needs.

MY BRAIN GETS TIRED QUICKLY AND I CAN GET DISTRACTED WHEN MY BRAIN IS TIRED. Encourage me to take brain breaks and help me set timers for these breaks. Brain breaks help me stay focused.

I AM VERY PASSIONATE ABOUT CERTAIN TOPICS AND ENJOY TALKING ABOUT THEM AT LENGTH.

You can help me by listening when you can, and let me know when you don't have the time to listen. You can say, "I am really interested in what you have to say, but I don't have the time now. Let's find a time when I am free and can listen to you." This validates my feelings and helps me understand that my thoughts are important. This will help reduce my anxiety and calm me down.

IF YOU SEE ME MAKING REPETITIVE MOVEMENTS WITH MY BODY, PLEASE DON'T GET SCARED.

I use repeated body movements like rocking, flapping, clapping, etc., to express my emotions, especially to indicate that I am excited.

I HAVE MY STRENGTHS AND WEAKNESSES, PLEASE APPRECIATE AND ENGAGE MY STRENGTHS TOO.

Even though my weaknesses might catch your attention, please try to identify my strengths as well. I engage well when you identify and utilize my strengths like coding, music, math, etc., to teach me.

THANK YOU FOR READING,
YOURS SINCERELY,
D.

ASD and ADHD can present unique challenges in a general education classroom. Traditional methods of teaching and classroom management may not always meet the needs of such children. D's letter to his teacher helps us understand the way children with ASD perceive and interact with the world around them. D's letter reminds us that, with appropriate support, children like D can thrive socially and academically in general education classrooms. Teachers need to ensure that such children's strengths are not overshadowed by their challenges. Teachers play an important role, not just in the educational journey of children on the spectrum, but also in making them feel heard, valued, and understood.

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Sapna Bhat is a speech language therapist who consults with different international schools in Greater Copenhagen in Denmark. She has a PhD from the All India institute of speech and hearing, Mysore and has 24 years of experience working as an educator, researcher, and a clinician in the field of speech and language therapy. She is passionate about neuro-inclusion and works closely with special educators and families to facilitate inclusive education and scaffold the social world for neurodivergent children.

From the 'world' to the 'word'

A Freirian look at reading

Amruta Veer and Swati Jain

Introducing children to reading is a complex and demanding task for parents and teachers. There have been long-standing ideological debates over the most effective methods to teach reading to young children. The 'reading wars', as they were called, oscillated between the phonics[1] method and the whole language approach.[2] Today, these debates have narrowed down to a balanced and integrated approach to literacy along with an emphasis on the context of the text and the direct inclusion of aspects from the child's surroundings.

The introduction of this approach to reading instruction raises important questions for educators. What is the importance of a child's surroundings and the context of the text in acquiring reading skills? How do the surroundings affect comprehension of the text and how does the comprehension interact with the child's existing knowledge?

We looked at Freire's seminal text, *The Importance of the Act of Reading* (1983), which discusses the nature of reading and its central role in literacy development. Freire situated reading within the broader context of understanding the world before decoding the text for meaning.

"Reading the world precedes reading the word. And reading the word implies continually reading the world." (Freire, 1983, p. 10)

In his view, the child's surroundings, experiences and observations form their 'world' or the context. The language of the adults around the child, their ideas, beliefs, and various aspects of daily life also contribute to it. The child should understand and decipher this 'world' before they try to decode

1. A science-based method where students sound out words to understand the text. Phonics help students read unfamiliar words on their own.
2. The whole language approach believes learning to read should be as natural as learning to speak. Children learn to read by guessing unfamiliar words and using written clues to figure them out.

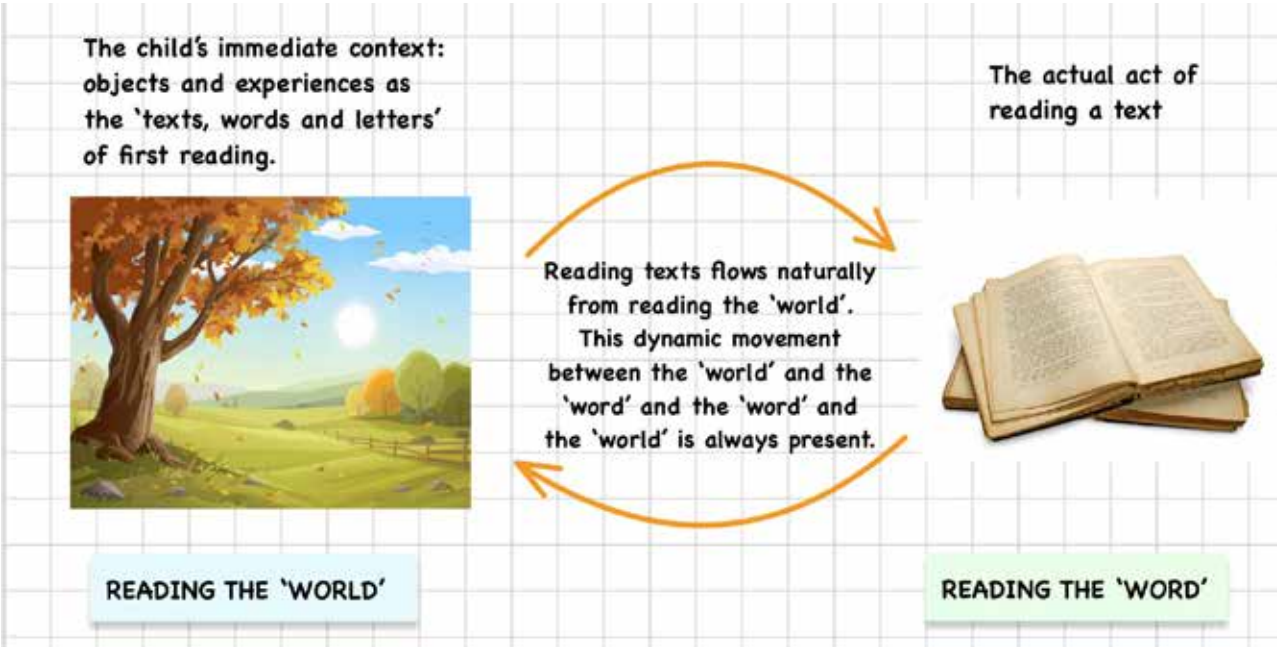

Figure 1: Dynamicity between the 'world' and the 'Word' (Adapted from Freire, 1983).

Images courtesy: Amruta Veer and Swati Jain

written texts. However, this is not a linear process (Fig. 1, see on previous page). The subsequent reading of the word cannot dispense with continually reading the world. He explains, *"Reading is not exhausted merely by decoding the written word or written language, but rather anticipated by and extending into the knowledge of the world" (Ibid.)*

As the word is decoded, its meaning expands the child's understanding of the world, which had at the outset, helped her to decode it. Thus, reading the context and reading the text are intertwined.

The physical context and surroundings of the child influence the meaning a word holds for the child. For example, the word 'tree' can hold different meanings for children raised in a village versus those raised in a city. Both children perceive trees in their surroundings, see how people interact with them, and hear the word used by them. This forms their unique context for understanding 'tree'. When the child encounters the word 'tree' in written language, she will not only decode the word but also understand its meaning pertaining to her individual context. For a child from a rural background, the tree might represent a place to play, climb, or find shade. She may even understand trees as important for farming, or for firewood, making trees an organic, integral part of their daily lives.

Whereas for a child from an urban background, a 'tree' would mean something limited and special, mostly confined to parks and sidewalks. She would hear about trees more as a part of a science lesson or stories, viewing it as a concept rather than a regular, familiar aspect of their lives. Both these children will decode the word 'tree' and construct their understanding based on what the tree signifies to them, in their individual contexts.

Freire proposes that establishing this relationship between the context and the text is crucial for reading (Fig. 2). Through this process, the word shaped by the reader's individual context, acquires a unique meaning for her and truly becomes 'hers. 'Freire explains this process as 'rewriting' the word, as now it possesses an individual or personal significance for the reader.

These three processes – perception, interpretation of the written word, and its 'rewriting' in relation to the context form the basis of critical reading. Critical reading should alter and expand the child's perception of the world. The child learns and

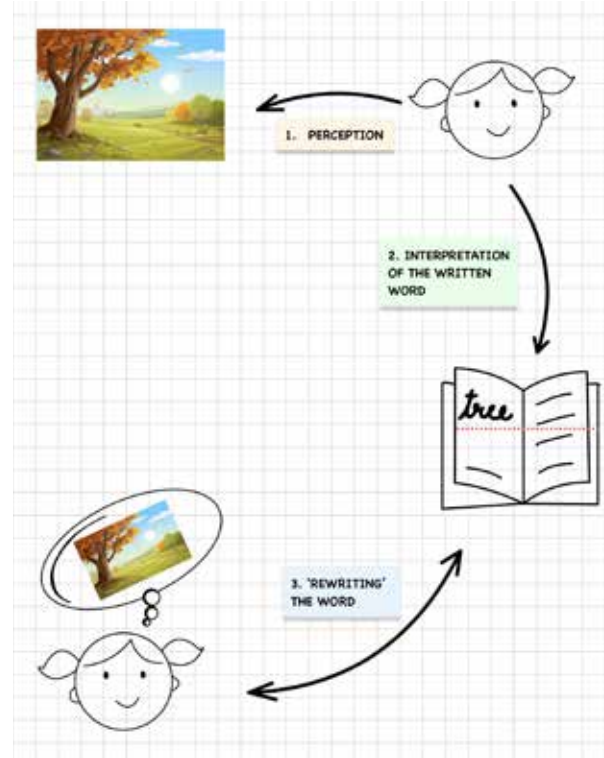

Figure 2: Critical Reading: Making the text ours (Adapted from Freire, 1983).

evolves through this interaction. Freire further argues that reading is 'the act of knowing' and 'the act of creativity'.

It is the creative task of the child to understand how the word 'tree' is constructed and learn to recognize, decode, and eventually write it.

Implications for parents and educators

Freire's article compels us to think about the approaches to reading taken by schools which primarily focus on instruction and practice of mechanics of reading, neglecting the importance of the real-world context and experiences. He challenged the mechanical and reductive view of reading which advocates for phonics – first instruction before addressing critical comprehension. He considers teaching literacy through exercises like memorizing vowel sounds, akin to filling the 'supposedly empty heads of learners' with the teacher's words. According to him, children's curiosity should be fostered so that they find the 'words' embodied in objects, ideas, and signs around them and feel a sense of discovery in finding these words in written text.

Pre-school is the time when the child is mediating the school world and the home world. The pedagogy should help foster a connection between these worlds. *'[With his teacher], reading the word never*

entailed a break with reading the world' (Freire, 1983, p. 8). The key to maintaining this continuity is by encouraging the child to speak with spontaneity about experiences, learning to read texts, and writing words that come from a child's ideas and observations and not from a textbook or the teacher's 'world'. This can be achieved by letting the child's everyday reality seep into the teaching process. It can form the basis of the context required to look at the written word meaningfully as well as nudge the child towards learning the formal rules of reading while keeping the spontaneity of the process alive. Educators and parents should be mindful to further this exploration so the child's journey toward literacy is rooted in their everyday actions and musings, while maintaining an active dialogue with the child about these experiences. In this way, their journey from the 'world' to the 'word' becomes a more organic one.

Freire also emphasizes that the vocabulary used in a literacy program must come from the learners' own language universe, reflecting their real language, anxieties, fears, needs, and aspirations. Surveying this 'word universe' provides words that are deeply connected to their lived experiences. This has significant implications for teaching children from marginalized communities, who often do not encounter their home language in the school setting. In a similar vein, Sylvia Ashton Warner, in the 1970s, developed a literacy method for teaching Māori children in New Zealand. Instead of relying on prescribed texts, learners were encouraged to engage with written culture on their own terms. They were taught words necessary to their lives and encouraged to write sentences and stories to share with other learners. *'No time is too long spent talking to a child to find out his key words, the key that unlocks himself, for in them is the secret of reading, the realisation that words can have intense meaning. Words having no emotional significance to him, no instinctive meaning, could be an imposition, doing him more harm than not teaching him at all. They may teach him that words mean nothing, and that reading is undesirable.' (Ashton-Warner, 1972, p.44)*

In this context, it is also essential to reflect on the true meaning of being "literate." The Ministry of Education (MOE) has defined it as the ability to read, write, and compute with comprehension (Porecha, 2024). However, is it adequate to simply teach children to sign their names and accurately read and write simple texts? Or should we aim for a more comprehensive definition of literacy – where children comprehend what they read and write, recognize its significance in their lives, and are equipped to critically engage with and navigate the world of written texts? (Menon, 2015)

Hellen Keller's (2012) account of learning the connection between the 'word' – water and 'the cool, wonderful something that was flowing over her hand' reveals very poignantly how the mystery of language unfolded before her. The joy and hope which came from this 'living word' is palpable in this autobiographical text.

"Everything had a name and each name gave birth to a new thought. As we returned to the house, every object which I touched seemed to quiver with life. That was because I saw everything with the strange, new sight that had come to me" (Keller, 2012).

This serves as an important reminder of the significance of the relationship between words, written texts, and the external world for a child. Learning to read brings this connection to life. As educators, it is crucial to recognize and strengthen this connection in teaching practices.

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The Constitution and General Will of people

Prakash Iyer

Ira: After our last conversation, I have been thinking about quite a few things regarding the Constitution and how we relate to it. I am still not able to wrap my head around one thing. As many as 300 people from different parts of the country came together and drafted a constitution that would constitute India as a democratic republic. How was it decided that we would be a democracy and not something else? How did these 300 people conclude that the rest of India wanted to be a democracy, when most of the population did not even know what it meant? Most importantly, they also decided on universal adult franchise!

Anubhuti: The process of drafting the Constitution itself was democratic and it was based on how the Indian society is and what it ought to be. During the debate within the Constituent Assembly, most of the members who held positions in the government were aware of the happenings in the nation. We also need to remember that the Constitution was being drafted in the midst of post-partition violence and displacement, and the concomitant effect on the way people think. They were receiving numerous letters and communication from people about the happenings in society and what ought to be done. The Constituent Assembly considered all these letters as the Will of the people. People wanted a society where such things would not happen. This required the Constituent Assembly to hypothesize the kind of nation required and have strong justifications for the same.

Ira: This almost sounds like a referendum, when the entire country's opinion is taken into account. But I am sure many people thought otherwise. You said even within the Constituent Assembly there were disagreements, and those disagreements were voted out.

Anubhuti: Yes, Ira. Some opinions were voted out. I am sure many individual assembly members were not very happy. For instance, Maharani Amrit Kaur was not pleased with the way the right to freedom of

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religion was treated[1]. Some people even thought that the definition of untouchability should be limited to a religion, rather than across religions.

My point is there were disagreements on both relevant and not-so-good issues. The Assembly still had to ensure that at least a minimum number of rights and duties were enforced and that the Constitution was coherent. You could not have conflicts within the Constitution.

This reminds me of the concept of the General Will proposed by Rousseau in his book *The Social Contract*.

Ira: General Will? How is that different from the personal wills of individuals?

Anubhuti: Rousseau said humans were basically free and had their own wills, wants, and desires. Everyone would want to live their life on their own terms. But humans also live together in large groups, and their lives do depend on others in the society. Therefore, they could not claim to live a life that would be detrimental to the larger society or do something that others in the society would not care for or benefit from. Rousseau said that individuals would see the benefits they were getting from living in a society, and so they would cherish others in the society and would not want to go against the common ways of living. He called the will of the society the General Will, which, in a way, is the product of numerous particular wills of individuals. While they are both different, he said that they cannot be in contradiction to each other.

When the Constituent Assembly was hypothesizing what individuals in our society would want, they were actually pondering over the General Will. So, any particular suggestion made either by members of the Assembly or by citizens through their letters, would be judged against the General Will.

When a person complains about something to the government (or Constituent Assembly, which we are discussing) about something, they are saying this is how our society is organized, and we want the society to be organized differently. They are not saying we do not want these other people who we are in conflict with to live here. Rather, they are saying, find some way of organizing the society, so that problems do not crop up.

Ira: You mean to say that when people were posing societal problems in their letters to the government, they were actually pointing out how law and society should be? And this was considered as additional inputs from the society?

Anubhuti: Yes. They may not have used the terms 'General Will' or 'Social Contract', but the Constituent Assembly identified these general issues with the rights that need to be defined, the duties of citizens, and the way the government was structured and what in the structure needed to be changed. But the change should be such that it did not value the interests of some people over those of others, and yet did not allow injustice to happen. So, for instance, even if some people said there was nothing wrong with caste discrimination, the Constituent Assembly disagreed. They valued the General Will and imagined the way India ought to be more than what some particular people thought.

It was real hard work and application of logic.

Ira: But I am still confused. How did so many people at that time know that the Constitution was democratic, and it is a better way of organizing society than any other? I mean, finally a small percentage of people made these decisions.

Anubhuti: Yes, of course, every citizen did not know democracy or could not even read the Constitution. Like Madhav Khosla famously said in his book[2], the Constitution was not only a means to dictate or prohibit actions but was also conceptualized as a pedagogical tool in order to build a new civic culture.

Over time, reconstituting India in this manner educated people in the democratic processes and the idea of how democratic societies functioned; in short, the Constitution educated us people into being democratic.

Ira: The first step would have been to educate people to vote for the right representatives and ensure they are doing the right things.

1. Episode 5 of Samvidhan TV series. https://www.youtube.com/watch?v=0U9KDQnIsNk&t=33s
2. Introduction of "India's Founding Moment: The Constitution of a Most Surprising Democracy" by Madhav Khosla

Anubhuti: Not exactly. Voting, of course, was and continues to be the most significant aspect of democracy. That is primarily because we chose the model of representative democracy. But, it is interesting that Rousseau disagreed with the idea of representation.

"One cannot transfer one's will to another, to do with as he or she sees fit, as one does in representative democracies. Rather, the General Will depends on the coming together periodically of the entire democratic body, each and every citizen, to decide collectively, and with at least near unanimity, how to live together, i.e., what laws to enact."[3]

He advocated direct democracy or republicanism as it is called. Every decision is made collectively by all people, like in a referendum. Whenever a decision has to be taken, each person would have to vote and proffer their opinion on why they voted for one decision over another.

Ira: How come we call India a republic when most citizens are active only in terms of voting?

Anubhuti: That is not true. Active citizenship is demonstrated by us in various ways. Every time a significant policy is made, public acceptance is asked for. Remember a draft of the new education policy was up for review for over a month, during which an online form collected responses from all of us and changes were made to the policy before it was finalized? Similarly, for other decisions, public consultations are organized. I attended one such consultation when a decision had to be taken on whether genetically modified Brinjal seeds should be bought by the nation or not. Farmers, activists, companies, and scientists participated in this debate. And the government decided not to introduce BT-Brinjal, as it was called. Another form of active citizenship is protests. Protests are a common form of expressing our dissent against some policy or law, and the government has to reconsider policies or laws, taking into account the general will of the nation. For instance, the farm laws meant to liberalize the agricultural sector were repealed after large scale protests from farmers. All protests by groups of people may not result in policy changes, but the point of protests is active citizenship and to give voice to the General Will of the people.

In all these situations, the government has to consider the General Will of the people, and not the opinions of particular groups.

Ira: You are saying India is constituted as a democratic republic, and as of today most people have been educated into democracy by the Constitution?

Anubhuti: Of course, we all are. I would add that this knowledge and understanding is not limited to educated and literate people like us, who can read the Constitution. Most of the citizenry are aware of their rights and duties and actively demonstrate that.

I am reminded of the last civics class when a student asked me what being *constituted as a democracy* meant. It was tough answering that question. I found the meaning of the word in a dictionary, which says, *"The structure, composition, physical makeup, or nature of something"* very useful.

When we use this definition in relation to the body and mind, we are referring to our overall health. When we use this for a society, we mean the nature of the society. India was a British colony until 1947 and we transformed into a democratic republic in 1950.

You are right in that it seems like a gamble to form a democratic state that demands so much from citizens most of who were not *literate or educated* in 1950. But the Constituent Assembly was right in accepting that the Constitution would serve as a pedagogical tool to educate us into a democratic culture. Our parents and grandparents seem to have done a good job, don't you think?

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3. Entry on "Social Contract Theory" in the Internet Encyclopedia of Philosophy. https://iep.utm.edu/soc-cont/

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